

# MAXIMISING EXAM MARKS IN **Senior maths**

ADVANCED Book 1

*Top tips for  
avoiding  
common errors!*



**BARBARA LYNCH**



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Barbara Lynch is former Head of Wesley College in Glen Waverley, Victoria, author of over 90 titles and now a consultant in areas such as strategic planning, governance, policy development, curriculum, assessment, reporting, CD-ROM and Internet publishing. Barbara has a long and enviable background in secondary education and mathematics.

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Clayton South VIC 3169  
Ph: +61 3 9558 4433  
[info@blake.com.au](mailto:info@blake.com.au)  
[www.blake.com.au](http://www.blake.com.au)

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# Introduction

Every year, examiners, teachers and assessors in every state produce detailed reports on how students have coped with the various questions on their exams. And every year the same errors are made by students. *Maximising Exam Marks in Senior Maths* is a collection of reproducible worksheets which are designed to assist senior students in those areas that they have traditionally found difficult.



Questions focusing on the common errors in the context of the topic area in which they typically occur will help students avoid wasting valuable marks. Key labels mark important content, concepts and techniques:

**THINGS TO KNOW** – Content that students should learn and be able to apply

**HINTS** – Helping students approach questions correctly

**TRICKS** – Giving short-cuts or easier ways to attack questions

**SLIPS** – Providing opportunities for students to practise working carefully to avoid common errors

**TRAPS** – Providing the types of contexts in which students make common errors

The table of contents below shows the topic areas used in each activity. Pages which stress appropriate examination and revision techniques are also included. The worksheets could be integrated into the learning process or used as part of examination or test revision. In most cases, an electronic calculator would be useful and where appropriate, a graphing calculator can be used.

The CD-ROM attached to the inside back cover contains a PDF version of this book.

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# 1 – Some stuff is assumed knowledge

There is some theory that you will not be retaught because it is assumed that you already understand it. For example, you may need to understand interval notation or alternatively how domains can be restricted using inequality notation. If the Binomial Theorem is required in your syllabus, you can use that instead of multiplying brackets out by hand. Some calculators can be used to factorise and expand expressions or to check your working.



## THINGS TO KNOW

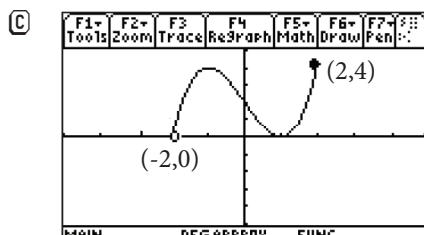
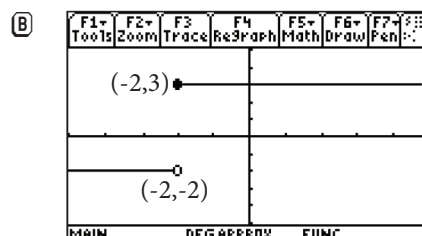
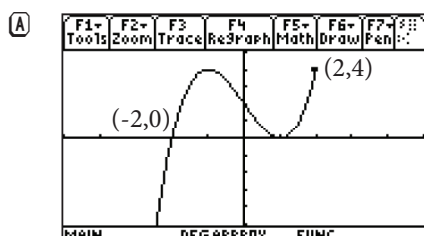
$x \in (a,b)$  can be written as  $a < x < b$

$x \in [a,b]$  can be written as  $a \leq x \leq b$

$x \in (a,b]$  can be written as  $a < x \leq b$

$x \in [a,b)$  can be written as  $a \leq x < b$

- ① What are the domains of the functions with the following graphs?



- ② If  $x^2 + y^2 = 16$  make  $y$  the subject of the equation.

- ③ Solve for  $x$ :  $\left(\frac{1}{3}\right)^x = 9$

**SLIP**

Be careful of reciprocals.

- ④ Make  $y$  the subject of  $\frac{1}{x} = \frac{1}{x^2} + \frac{1}{y}$ .

- ⑤ Solve for  $x$ :  $\sqrt{x+12} = x$ .

**TRAP**

When you square both sides of equations, substitute back to check there are no invalid solutions!

- ⑥ Copy and complete this sum to give a whole number answer:

$$\sum_{n=4}^7 (2n-1) = 2 \times 4 - 1 + (2 \times 5 - 1) + \underline{\hspace{2cm}} + \underline{\hspace{2cm}} =$$

**HINT**

Know the rules and the meanings of symbols!



## THINGS TO KNOW

These may be on your formula sheet but you should still know the rules so you recognise them when they come up.

Arithmetic series:  $a + (a + d) + \dots + a + (n - 1)d = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + \ell)$

Geometric series:  $a + ar + ar^2 + \dots + ar^{n-1} = a \frac{(1 - r^n)}{1 - r}; r \neq 1$

$S_{\infty} = \frac{a}{1 - r}; 0 < |r| < 1$

7 For  $1 + 2 + 3 + \dots$  find an expression for:

A the  $n$ th term.

B the series.

What happens to the series as  $n$  approaches infinity?



### TRAP

Do not confuse the series with the term of a sequence. Think carefully about what is happening as  $n$  gets large.

8 For  $1 + \frac{2}{3} + \frac{4}{9} + \dots$  find an expression for:

A the  $n$ th term.

B the series.

What happens to the series as  $n$  approaches infinity?

9 Expand  $(2b - 3)^3$ .

10 Expand  $(9x - 9)^2$  by taking out a common factor first.

11 Find the term in  $x$  of the expansion  $(3x - 2y^2)^4 = 81x^4 + 4 \times 27x^3 \times (-2y^2) + 6 \times (3x)^2 \times (-2y^2)^2 + \dots$

12 Use the binomial theorem to find the term in  $x$

in the expansion of  $(3x - \frac{1}{x^2})^7$ .

13 Divide  $6a^3 + 3a^2 + 2a - 1$  by  $3a + 2$ .



### TRAP

9 is squared too!



### SLIP

- Be careful with the indices, brackets and negative sign.
- Notice that you are asked for the term, not the number.

## HINT



What method will you use? Long division? Synthetic division? Fractions?

## 2 – Best ways to factorise

Factorising is writing an expression as a product of its factors. It is one way of simplifying an expression and is a necessary skill in the solution of many mathematical problems. Factorising over  $\mathbb{Q}$  gives factors containing only rational numbers while factorising over  $\mathbb{R}$  gives factors containing irrational numbers as well. Expanding usually means multiplying out brackets. The main thing is to be careful of slips in your work and practise, practise, practise.



### THINGS TO KNOW

Common factor first – always!

#### Two terms:

Difference of perfect squares:  $a^2 - b^2 = (a - b)(a + b)$

Difference of perfect cubes:  $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Sum of perfect cubes:  $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

#### Three terms:

☞ Perfect square:  $a^2 + 2ab + b^2 = (a + b)^2$

$$a^2 - 2ab + b^2 = (a - b)^2$$

☞ Quadratic trinomial using cross method or splitting the middle term for the distributive law in reverse

☞ Quadratic trinomial using completing the square

☞ Quadratic trinomial using substitution to make it a simpler quadratic trinomial

☞ Quadratic trinomial using completing the square with the middle term and then using the difference of two squares

☞ Quadratic trinomial using the factor theorem

☞ Quadratic trinomial using the quadratic formula

#### More than three terms:

☞ Group terms two and two – so that there is a factor common to each pair

☞ Group terms three and one – where the three terms form a perfect square and so there is the difference of perfect squares

☞ Factor theorem (the factors of the constant help find the factors)

**Hint:** You can check your factors are right by multiplying them out.

① Factorise, where possible, each of the following over the real field,  $\mathbb{R}$ :

A  $3x + 6y - 27z$

B  $100p^2 - 4q^2$

C  $a^2 - 15$

D  $8x^3 + 64y^3$

E  $x^2 + y^2$

### TRICK

Any common factors should be taken out before any other method is applied.

② Factorise each of the following over the real field,  $\mathbb{R}$ :

A  $4x^2 - 12x + 9$

B  $2y^2 + 3y - 9$

C  $x^2 - 8x + 4$

D  $a^4 - 12a^2 + 35$

E  $b^3 - 4b + 15$

### THINGS TO KNOW

The general quadratic expression  $ax^2 + bx + c$  has factors given by:

$$ax^2 + bx + c = a \left( x + \frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a} \right) \left( x + \frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a} \right)$$



## 2 – Best ways to factorise (cont'd)

- ③ Can the expression  $2a^2 + 5a + 3$  be factorised over the real field? Give reasons for your answer. If it factorises, what sort of factors does it have?

- ④ Factorise over the real field,  $R$ :

- A  $b^3 - 64$   
B  $3x^2 - 5x - 3$   
C  $x^2 + 4xy + 4y^2 - 9z^2$   
D  $y^3 + 2y^2 - 11y - 12$

### TRICK

If you are using the factor theorem, the factors of the constant term help you find the factors to test for.

- ⑤ If  $3y^3 + 8y^2 - 20y - 16$  can be written in the form  $(y - 2)(ay^2 + by + c)$ , find  $a$ ,  $b$  and  $c$  and hence find the linear factors of the polynomial over the rational field.

### TRAP

Answer the question and use 'hence'.

- ⑥ Find linear factors of  $pr^3 - qr$  over the real field, where  $p, q \in R^+$ .

### HINT

These don't have numbers in them. Think of the  $p$  and  $q$  as if they were numbers and  $r$  as the variable and then look at the key words:

- ☞ 'linear' (from line graph) means that there are no squares, cubes, etc. of  $r$  in the answer
- ☞ real field and  $p$  and  $q$  are real positive numbers, which means that you can have roots in the answer.

- ⑦ A Find the quotient and remainder when  $x^3 - 1$  is divided by  $x^2 - 1$  and state your answer clearly.

### HINT

Long division is okay but if you are using synthetic long division, be careful that you can interpret the tables to find the remainder.

- B For what two values of  $k$  is  $x + 1$  a factor of  $P(x) = x^n + k$  if  $n$  is a positive integer?

- C If  $x^2 - 1$  is a factor of  $P(x) = x^n + k$ , find all possible values of  $n$  and  $k$ .

### TRAP

Make sure that your answers to parts b and c are consistent.

# 3 – Good algebra skills are a 'must'

Algebra appears in some form or another in almost every part of Mathematics. This means that you need to be able to work through any question without making careless mistakes. For a start, make sure you copy any equations carefully from the paper. Show all your working so that you are explaining your reasoning and intentions. Also, use graphs and drawings to help with your explanations. Be careful to use the correct notation rather than being sloppy about it. Number all the questions and their parts clearly and watch out for words such as 'hence' and instructions such as 'use the factor theorem'. Finally, remember that the number of marks is usually proportional to the amount of work you are expected to do so don't spend a long time on a question worth 1 or 2 marks.



## TRAP

① Find  $x$  if  $x^2 + h^2 = 16$ .

② Simplify  $\frac{e^2}{8} - \frac{1}{8}\left(\frac{e^2}{8} - \frac{1}{8}\right)$ .

③ Evaluate  $\frac{1}{9} \frac{(2x^2 + 1)^{\frac{5}{2}}}{\frac{9}{2}}$  when  $x = 2$ .

④ Show that  $\left(\frac{x+1}{x}\right)^4 = -4$  can be written as  $5x^4 + 4x^3 + 6x^2 + 4x + 1 = 0$ .

⑤ If  $x = 100t$  and  $y = -4.9t^2 + 100t + 4000$ , find  $y$  in terms of  $x$ .

⑥ Find  $a$  and  $b$  if  $2ae^{2x} \cos x - 6e^{2x} \sin x = be^{2x} \sin x + 4e^{2x} \cos x$ .

⑦ Show that  $-3(-x-1)(3-2x)^{\frac{1}{2}} - (3-2x)^{\frac{3}{2}} = 5x\sqrt{3-2x}$ .

$4 - h$  is not one of the two possible answers.

## SLIP

Plenty of opportunities to make slips in this one.

## TRICK

Equate coefficients.

## SLIP

Be careful of the negatives. Also remember the meaning of the fractional powers and the fact that  $(3-2x)^{\frac{3}{2}} = (3-2x)(3-2x)^{\frac{1}{2}}$ .

⑧ Given that  $\frac{5x-2}{(x^2+2)(2x+1)}$  can be written in the form  $\frac{x+A}{x^2+2} + \frac{B}{2x+1}$ , find the values of  $A$  and  $B$ .

⑨ By considering  $f(x) = 4 \log_e x - x$ , show that the curve with equation  $y = 4 \log_e x$  meets the line with equation  $y = x$  at a point  $P$  whose  $x$ -coordinate lies between 1 and 2.

## HINT

You must use the same form of the fractions as the one given rather than your own. Algebraic errors can be minimised if you use brackets carefully. Equating coefficients is a helpful method.

## HINT

The easiest method is to look at the sign of the function at  $x = 1$  and at  $x = 2$  since the graph of the function crosses the  $x$ -axis if the function is negative on one side and positive on the other. Be careful of trying to solve inequations like  $f(x) < 0$  or trying to use exponential forms as this can make it a very difficult problem that is hard to reason. In its original form this question was worth only 1 mark so that would have told you that it was going to have a simple method.

⑩ Expand and simplify  $\left\{3x^{\frac{1}{2}}(4-x^2)^{\frac{1}{4}} + \frac{1}{(4-x^2)^{\frac{1}{4}}}\right\}^2$ .

⑪ If  $\left(\frac{x+y+z}{3\sqrt[3]{xyz}}\right)^3 \geq \left(\frac{x+y}{2\sqrt{xy}}\right)$ , show that if  $\left(\frac{x+y}{2}\right) \geq \sqrt{xy}$  then  $\frac{x+y+z}{3} \geq \sqrt[3]{xyz}$ .

## TRICK

Remember the middle term and make sure you simplify your answer.

⑫ Show that if  $a, b, c$  and  $d$  are constants,  $x = \frac{b}{c} \log_e \left(\frac{2ab}{2ab - cd}\right)$  satisfies the simultaneous equations:

$$y = bx - \frac{b}{c}(b-a)\left(1 - e^{-\frac{cx}{b}}\right) \dots \dots \dots (1)$$

$$y = d + bx - \frac{b}{c}(b+a)\left(1 - e^{-\frac{cx}{b}}\right) \dots \dots \dots (2)$$

## TRAP

This just takes careful Algebra because the manipulations are a little complicated and there is plenty of room for error, particularly with signs when the brackets in equations (1) and (2) are multiplied out before subtracting.

# 4 – De Moivre's Theorem

This theorem makes it easier to find products, sums, powers and roots of complex numbers written in polar form. Of course the correct 'cis' form is essential so you need to be careful that you have your angles in the correct quadrants. Also, because  $i^2 = -1$ , you need to watch the positive and negative signs in equations and calculations. Some of the theory may be on your formula sheet but you should still know the rules so you recognise them when they come up.



## THINGS TO KNOW

De Moivre's Theorem: If  $z = r \operatorname{cis} \theta$  then  $z^n = r^n \operatorname{cis} n\theta$  or  $z^n = r^n [\cos(n\theta) + i \sin(n\theta)]$  for  $n \in \mathbb{J}$  and  $n \in \mathbb{Q}$  and if  $w = r_1 \operatorname{cis} \theta_1$  and  $z = r_2 \operatorname{cis} \theta_2$  then  $w \cdot z = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2)$  and  $\frac{w}{z} = \frac{r_1}{r_2} \operatorname{cis}(\theta_1 - \theta_2)$ .

- ① Solve for  $z$ :  $z^3 + 1 = 0$  and give your answers in the form  $r \operatorname{cis} \theta$ .

## HINT

Finding the roots of a negative number is a useful technique to learn. Note that the question says 'answers' not 'answer', so that is a clue.

## TRAP

Note that the question asks for an integer value; the answer is not  $\frac{3}{2}$ .

- ② If  $z = 2 \operatorname{cis} \frac{2\pi}{3}$ , find the least integer value of  $n$  for which  $z^n = 1$ .

- ③ Solve the equation  $z^4 = -4$  for  $z$  using de Moivre's Theorem. Give your answer in the form  $z_n = x_n + iy_n$  where  $n = 1, 2, 3, 4$ . Do the answers form conjugate pairs?

- ④ If  $f(z) = z^2$ , calculate  $f(\operatorname{cis} \frac{3\pi}{4})$ ,  $f(\operatorname{cis} \frac{3\pi}{2})$ ,  $f(\operatorname{cis} \frac{9\pi}{4})$  and  $f(\operatorname{cis} 3\pi)$ .

## TRICK

Make sure you simplify at each stage.

## TRICK

Try writing  $-4$  in the form  $r \cos \theta + i r \sin \theta$ , otherwise the question will be quite difficult. Remember to give your answers in the correct form:  $x + iy$ , as requested. This also helps you to see if they are conjugate pairs. Conjugate pairs have the form  $a + ib$  and  $a - ib$ .

- ⑤ Point A represents  $z = r \operatorname{cis} \theta$  where  $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ . The other two points represent  $z^2$  and  $z^4$ .

A Which of A, B and C represents  $z^2$ ?

B Which of A, B and C represents  $z^4$ ?

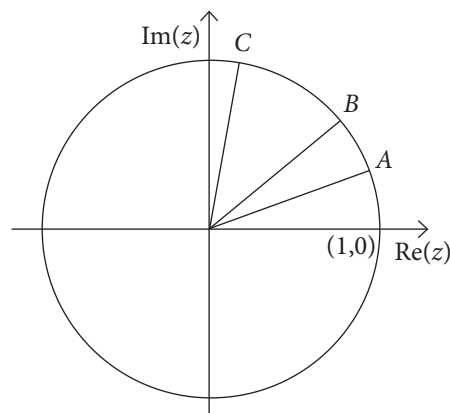
C On the same diagram, show the complex number  $z^2 + z^4$ .

D Hence or otherwise show that

①  $\operatorname{Arg}(z^2 + z^4) = 3\theta$ .

②  $|z^2 + z^4| = 2 \cos \theta$ .

E Explain briefly why  $z^2 + z^4 = 2 \cos \theta \operatorname{cis} 3\theta$  using the results of part d.



## THINGS TO KNOW

Based on the complex number and trigonometry theory that you need to learn:

$\sin A + \sin B = 2 \sin \frac{1}{2}(A+B) \cos \frac{1}{2}(A-B)$

$\cos A + \cos B = 2 \cos \frac{1}{2}(A+B) \cos \frac{1}{2}(A-B)$



Ⓕ Hence:

Ⓘ find the exact values of  $\theta$ ,  $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ , for which  $z^2 + z^4$  is purely real.

Ⓙ find the values of  $z^2 + z^4$  for each value of  $\theta$ .

Ⓚ mark the results on an Argand diagram.

**TRAP**

Combining your knowledge of geometry, trigonometry and complex variables makes it harder.

Ⓖ Ⓐ If  $\alpha = \sqrt[3]{-1}$  where  $\alpha$  is not real, show that  $\alpha^2 - \alpha + 1 = 0$ .

Ⓑ Hence simplify  $(\alpha^2 + 1)^6$ .

Ⓙ If for complex number  $w$ ,  $\operatorname{Re}\left(\frac{1}{w-1}\right) = -\frac{1}{2}$ , what is  $\operatorname{Re}(w-1)$ ?

**HINT**

☞ There are two ways to do part a: either use factors  $\alpha^3 + 1 = (\alpha + 1)(\alpha^2 - \alpha + 1)$  or de Moivre's Theorem and  $r\operatorname{cis}\theta$  form.

☞ Remember that 'hence' means that you must use the previous result to get  $(\alpha^2 + 1)^6 = \alpha^6$ .

**SLIP**

The answer is **not** -2!

Ⓖ For  $n > 2$  and  $\omega$  is the  $n$ th roots of unity,  $\omega \neq 1$ , expand the left hand side to show that  $(\omega - 1)(1 + 2\omega + 3\omega^2 + 4\omega^3 + \dots + n\omega^{n-1}) = n$ .

**HINT**

You must say why at each step.

Ⓙ Ⓐ If  $z = \operatorname{cis}\frac{\pi}{8}$  and  $\cos^2\left(\frac{\pi}{8}\right) = \frac{2+\sqrt{2}}{4}$  and  $\sin^2\left(\frac{\pi}{8}\right) = \frac{2-\sqrt{2}}{4}$ , evaluate  $\left(\sqrt{\frac{2+\sqrt{2}}{2}} + \sqrt{\frac{2-\sqrt{2}}{2}}i\right)^6$ .

Give your answer in polar form.

Ⓑ Find  $\left(\sqrt{\frac{2-\sqrt{2}}{2}} + \sqrt{\frac{2-\sqrt{2}}{2}}i\right)^n$ .

ⓀⒶ If  $\alpha = 1 + i\sqrt{3}$  and  $\beta = 1 - i$ , find

Ⓘ  $\frac{\alpha}{\beta}$  in the form  $x + iy$ .

Ⓙ  $\alpha$  in modulus/argument form.

Ⓑ If  $\beta = \sqrt{2}\operatorname{cis}\left(\frac{\pi}{4}\right)$  in modulus/argument form, find  $\frac{\alpha}{\beta}$  in this form.

Ⓒ hence find the exact value of  $\cos\left(\frac{\pi}{12}\right)$ .

**SLIP**

There is plenty of scope for error if you are not careful.

☞ Watch the use of brackets at each step.

☞ Don't lose the denominator in the calculation.

☞ You will need to realise (make real) the denominator in part a ii.

☞ 'Hence' tells you to use previous results.

☞  $\cos\left(\frac{\pi}{12}\right) \neq \cos\left(\frac{\pi}{3}\right) - \cos\left(\frac{\pi}{4}\right)$

# 5 – Complex numbers and algebra

Know your standard rules for the Algebra of complex numbers and remember to keep in mind whether you are working with real numbers or complex numbers.



## THINGS TO KNOW

Polynomial equations over the complex field have the form

$$P(z) = a_n z^n + a_{n-1} z^{n-1} + a_{n-2} z^{n-2} + \dots + a_1 z + a_0 = 0; a_n \neq 0$$

where  $n$  is the degree of the polynomial and  $a_i$  are the coefficients of the unknown,  $z \in \mathbb{C}$ .

### The Fundamental Theorem of Algebra:

Every polynomial equation of degree  $n > 0$  with complex coefficients has  $n$  complex solutions (including repeated ones).

- ① Solve the quadratic equation  $z^2 + iz - 1$ , expressing your answers in exact Cartesian form.
- ② State how many solutions over the complex field and then solve.

Ⓐ  $(z - 2i)^2(z + 1) = 0$

Ⓑ  $z^3 = -1$

- ③ For the equation  $z^4 + (4 - 2i)z^3 + (3 - 8i)z^2 - (4 + 8i)z - 4 = 0$ ,

Ⓐ how many solutions are possible, including repeated ones?

Ⓑ if two of the solutions are  $-2$  and  $i$ , write down the other solutions.

## HINT

What is meant by exact Cartesian form? What is the easiest method of solving such a quadratic equation? Make sure you simplify your answer.

## THINGS TO KNOW

☞ If a polynomial  $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$  has real coefficients, then any complex zeros occur in conjugate pairs; that is, if  $a + bi$  is a zero then so is  $a - bi$  and vice versa.

☞ Notice that 'write down' means no calculations will be needed and you only need to find the other solutions.

- ④ The roots of the complex equation  $z^3 + az^2 + bz + c = 0$  are  $3i$  and  $2$ . If  $a, b, c \in \mathbb{R}$ , what is the third root?
- ⑤ Polynomial  $P(z) = z^4 + 4z^3 + 3z^2 + 2z + 2 = 0$  has one real root. Alice found the other three roots to be non-real, but Betty found the other three roots to comprise one real and two non-real. Which student was more likely to be correct?

## HINT

No calculation is required and you should only guess the answer as a last resort. This question uses the Fundamental Theorem of Algebra.

- ⑥ Show that the quadratic equation  $z^2 + (4 + 2i)z + (3 + 4i) = 0$  has 2 equal solutions (that is, 1 solution).

## HINT

Remember the discriminant is  $\Delta = b^2 - 4ac$ .



7 Let  $P(x) = x^4 - 5x^3 + 2x^2 + 22x - 20$ .

(A) Use the remainder theorem to show that  $x + 2$  is a factor of  $P(x)$ .

(B) Find  $Q(x)$  such that  $P(x) = (x + 2)Q(x)$ .

**TRAP**

There is plenty of opportunity to make careless mistakes in this question! When asked for the quadratic factor, you should multiply it out, not leave it in factor form. When you are asked for zeros then you can't just give the factors as answers. Also, of course, it's easy to make a mistake on the calculator when the numbers are a bit 'yukky'.

(C) The complex number  $w$  is a zero of  $Q(x)$  with  $w + \bar{w} = 6$  and  $w\bar{w} = 0$ . Write down a quadratic factor of  $Q(x)$ .

(D) Find the non-zero roots of  $Q(x)$ .

(E) Hence find all the zeros of the quartic polynomial,  $P(x)$ .

8 Factorise  $x^4 - 3x^2 - 4$

(A) over the real field.

(B) over the complex field.

9 Factorise  $x^2 + 3x + 1$  over the real field.

10 Factorise  $x^2 + 3x + 3$  over the complex field.

11  $P(z) = az^3 + bz^2 + cz + d$  can be factorised to  $(z - 2)(z + 1 + 4i)(z + 1 - 4i)$ . Expand the factors to find  $a, b, c$  and  $d$ .

12 How many negative real roots does the polynomial  $P(z) = 573z^4 - 892z^3 + 2z^2 + 573z - 89$  have?

13 If  $f(z) = z^2 + 4iz - 4$  and  $g(z) = z$ , then what is the value of  $f(g(3 + 2i))$ ?

**SLIP**

Be careful of your signs and how you group the complex terms. Remember to substitute each root accurately.

**HINT**

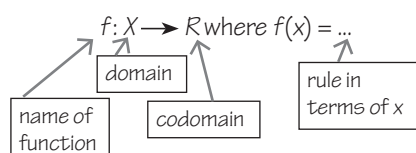
Look at factors of 89.

# 6 – Functions and relations

The theory of functions is fundamental to any study of Mathematics. You need to know the rules and definitions, have accurate Algebra and good graphing skills.



## THINGS TO KNOW



For an odd function  $f(-x) = -f(x)$  and for an even function  $f(-x) = f(x)$ .

The inverse function  $f^{-1}(x)$  is such that  $f^{-1}[f(x)] = x$ .

A function can be many-to-one (has no inverse function) or one-to-one (has inverse function).

- ① A Show that  $f(x) = \sin(x^3)$  is an odd function.
- B Hence state the area under the graph of  $y = f(x)$  between  $x = -\frac{\pi}{6}$  and  $x = \frac{\pi}{6}$ .

## HINT

Thinking about the graph may help.

- ② Consider the following statements about a polynomial  $P(x)$ :
- A If  $P(x)$  is even, then  $P'(x)$  is odd.
- B If  $P'(x)$  is even, then  $P(x)$  is odd.

Indicate whether each of these statements is true or false. Give reasons for your answers.

- ③ If  $f(x) = x^2 + 1$ , calculate  $f(0)$ ,  $f(1)$ ,  $f(2)$  and  $f(3)$  and state whether the function is increasing or decreasing in the interval  $[0, 3]$ .

## THINGS TO KNOW

Remember, the principal value of  $\sin^{-1}x$  is written  $\sin^{-1}x$  with domain:  $[-1, 1]$  and range:  $[-\frac{\pi}{2}, \frac{\pi}{2}]$  or  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ .

- ④ For the function  $f(x) = \sin^{-1}(\frac{x}{4})$ , find the inverse function  $f^{-1}(x)$ .

- ⑤ State the largest domain for which the following functions are defined:

A  $f(x) = x\sqrt[4]{4-x^2} + \frac{1}{\sqrt[4]{4-x^2}}$

B  $g(x) = \sqrt{x}\sqrt[4]{4-x^2} + \frac{1}{\sqrt[4]{4-x^2}}$

## TRAP

Be careful of the ends of the domain and think about what square roots and fourth roots need in order to exist.

- ⑥ If  $f(x) = \log_e(2 + 3e^x)$  for all  $x$ , show that  $f(x)$  has an inverse.

## TRAP

Sketching a graph of the function is not sufficient to 'show' since you must be certain there are no turning points. Remember  $f(-x)$  is not the inverse.

## HINT

You can interchange  $x$  and  $y$  to find the inverse and differentiate to show that the function is always increasing.



- 7 Indicate on the number plane the region satisfied by  $y \geq |x - 1|$ .

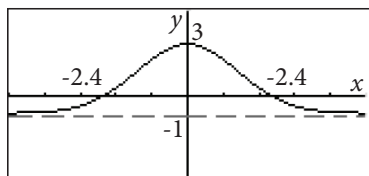
## HINT

Notice that it says 'number plane' not 'number line'. You need to get the correct shape and shading and show any intercepts. Make sure you test a point for the required region and show it clearly.

- 8 Given the sketch graph of  $y = f(x)$  shown, make a separate sketch of each of the following:

A  $y = (f(x))^2$

B  $y = \frac{1}{\sqrt{f(x)}}$



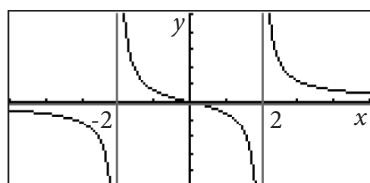
- 9 Given the sketch graph of  $y = f(x)$  shown, make separate sketches of each of the following:

A  $y = f(x + 2)$

B  $y = |f(x)|$

C  $y = \sqrt{f(x)}$

D  $y = f(|x|)$



## TRICK

- Remember that it is possible for a graph to pass through its own asymptote.
- Make sure you clearly show the difference between stationary points and cusps.
- Making your sketch large will help.

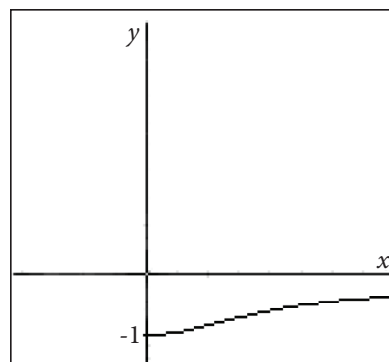
## TRICK

- Make sure that you show smooth functions and give the x-intercepts clearly.
- Show any transformations, reflections in particular.

- 10 The diagram shows the sketch graph of the function  $y = f(x)$

where  $f(x) = \frac{-4}{x^2 + 4}$  for  $x \geq 0$ .

- A On the same axes, sketch the graph of the inverse function  $y = f^{-1}(x)$ .
- B State the domain of  $f^{-1}(x)$ .
- C Find an expression for  $y = f^{-1}(x)$  in terms of  $x$ .
- D If the graph of  $y = f^{-1}(-x)$  meets the graph of  $y = f(-x)$  at exactly one point,  $P$ :



i write down the expressions for  $y = f^{-1}(-x)$  and  $y = -f(x)$ .

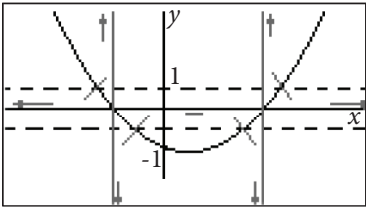
ii write down the equation and what needs to be solved to find the x-coordinate of  $P$ .

iii show by substitution that 0.85 is an approximate solution for the value of  $x$ .

## HINT

- You should understand the difference between  $f^{-1}(x)$ ,  $-f(x)$  and  $f(-x)$  and how their domains, ranges and graphs will compare.
- Any graph you draw should show the significant features clearly.
- Be careful in manipulating algebraic expressions with negatives and square roots.

# 7 – Graphs of functions and relations



When sketching the graphs of any function, you must make sure that you check the domains and ranges and label all the intercepts and asymptotes. You should make sure that the examiners see a distinction between your working graphs and your final answer.



## THINGS TO KNOW

### Reciprocal graphs

| Key                           | $x = 0$                                                                   | $y = 0$                  | $y = 1$                                               | $x = -1$                                                   | $f(x)$ increasing           | $f(x)$ decreasing           | $f(x)$ positive              | $f(x)$ negative              |
|-------------------------------|---------------------------------------------------------------------------|--------------------------|-------------------------------------------------------|------------------------------------------------------------|-----------------------------|-----------------------------|------------------------------|------------------------------|
| graph of $y = f(x)$           | gives $y$ -intercept                                                      | gives $x$ -intercept     | point on graph $f(x) = 1$                             | point on graph $f(x) = -1$                                 | graph has positive gradient | graph has negative gradient | graph is above the $x$ -axis | graph is below the $x$ -axis |
| graph of $y = \frac{1}{f(x)}$ | $y$ -intercept is $\frac{1}{f(0)}$ ; a horizontal asymptote if $f(0) = 0$ | gives vertical asymptote | $\frac{1}{f(x)}$ passes through same point $f(x) = 1$ | $y = \frac{1}{f(x)}$ passes through same point $f(x) = -1$ | graph has negative gradient | graph has positive gradient | graph is above the $x$ -axis | graph is below the $x$ -axis |

- 1 For the function with the rule  $y = \sec\left(\frac{\pi - 2x}{3}\right)$ :
- A state the period
- B use transformations to sketch one period of the graph.

- 2 Sketch the graph **over the principal domain** of:

- A  $y = 3\sin\left(\frac{x}{2}\right) + 1$
- B  $y = -\tan(2x - \pi)$
- C  $y = \cot\left(3x - \frac{\pi}{2}\right)$

### TRAP

Look at the bold type in the question. The principal domain is the domain for which the function is 1:1 and is defined as follows:

|           |                                 |                                 |              |
|-----------|---------------------------------|---------------------------------|--------------|
| Function: | $\sin^{-1}x$                    | $\cos^{-1}x$                    | $\tan^{-1}x$ |
| Domain:   | $[-1, 1]$ or $-1 \leq x \leq 1$ | $[-1, 1]$ or $-1 \leq x \leq 1$ | $\mathbb{R}$ |

- 3 For the function  $f(x) = \frac{2}{\log_e x}$ :
- A write down the domain
- B use reciprocal ordinates or otherwise to sketch the graph.

- 4 Sketch each of the following graphs.

- A  $y = e^{1-x}$
- B  $y = -e^{1-x}$
- C  $y = \sin 2x; -3 \leq x \leq 3$
- D  $y = e^{1-x} \sin 2x; -\pi \leq x \leq \pi$

### TRICK

You can use a calculator if you have one but be careful when you transfer the graph from the calculator screen to the paper.



5 For the function  $f(x) = \frac{4}{x^2 - 2x - 3}$ :

A show that  $f(x) = \frac{1}{x-3} - \frac{1}{x+1}$

B hence or otherwise sketch the graph of  $f$ .



**TRAP**

Check the range. Mark an extra point to show the shape.

6 For the function  $f(x) = \frac{x^2 - 4x + 4}{x - 2}$ :

A simplify the expression

B sketch the graph of  $f$ .



**TRAP**

Check the domain.

7 Sketch the graph with equation  $y = x + \frac{0.5x}{x^2 - 4}$  showing clearly any asymptotes and points where the graph meets the axes.

8 For the function  $f(x) = \frac{1}{x^2 - x + 1}$ :

A copy and complete the following table relating the values of the denominator and the values of the function.

| Denominator | 0 | <+1 | +1 | >+1 | -1 | maximum | minimum | $\rightarrow 0^+$ | $\rightarrow 0^-$ | $\rightarrow^+ \infty$ |
|-------------|---|-----|----|-----|----|---------|---------|-------------------|-------------------|------------------------|
| $f(x)$      |   |     |    |     |    |         |         |                   |                   |                        |

B hence use reciprocal of ordinates or otherwise to sketch the graph of  $f$ .



**TRAP**

Check the domain.  
Does the function exist for all  $x$ ?

9 Sketch the graph of the function with rule  $y = \frac{9x^2}{x^2 - 4}$  showing all asymptotes.

**TRICK**

Analyse the horizontal asymptote. Use facts like the graph passes through the origin and the function is even. Calculus is of little help. Perhaps you may like to use  $y = 9 + \frac{36}{x^2 - 4}$ .



10 The graph with equation  $y = \frac{1}{x^2 - Bx - C}$  has no vertical asymptotes. Find the relationship between real constants  $B$  and  $C$ .



**HINT**

Think about the number of  $x$ -intercepts the graph of  $y = x^2 - bx - c$  will have in this case.

# 8 – Coordinate geometry and conic sections

Many of the facts about conic sections may be given on the examination formula sheet. You should still know these facts so that you recognise them when they come up.



## THINGS TO KNOW

| Name                 | Cartesian equation                              | Foci          | Equations of directrices | Other relationships                                              |
|----------------------|-------------------------------------------------|---------------|--------------------------|------------------------------------------------------------------|
| Ellipse              | $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$         | $(\pm ae, 0)$ | $x = \pm \frac{a}{e}$    | $b^2 = a^2(1 - e^2), 0 < e < 1$                                  |
| Translated ellipse   | $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ |               |                          |                                                                  |
| Hyperbola            | $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$         | $(\pm ae, 0)$ | $x = \pm \frac{a}{e}$    | $b^2 = a^2(e^2 - 1), e > 1$<br>Asymptotes $y = \pm \frac{b}{a}x$ |
| Translated hyperbola | $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ |               |                          |                                                                  |
| Parabola             | $y^2 = 4ax$                                     | $(a, 0)$      | $x = -a$                 | $e = 1$                                                          |
| Translated parabola  | $(y-k)^2 = 4a(x-h)$                             |               |                          |                                                                  |

- ① For the points  $A(x, y)$  and  $B(4, -1)$ , the point  $P\left(\frac{2}{7}, \frac{7}{5}\right)$  divides the interval  $AB$  internally in the ratio 3:2. Find the coordinates of  $A$ .

**SLIP**

Be careful of your Algebra and make sure you use the correct formula.

- ② For the points  $A(2, -2)$  and  $B(8, 1)$ , find the coordinates of the point  $P$  which divides the interval  $AB$  **externally** in the ratio 7:2.

**TRAP**

Read the question carefully; the division is not internal! So point  $P$  is outside the interval  $AB$  and to the right of  $B$ . You may not have a formula to use.

- ③ Show that the equation of the ellipse for which the  $x$ -axis is a tangent at the point  $(2, 0)$  and the  $y$ -axis is a tangent to the same ellipse at  $(0, 1)$  is  $\frac{(x-2)^2}{4} + (y-1)^2 = 1$ .

**TRICK**

A sketch graph will help.

- ④ Find the minimum value of  $y$  reached by the ellipse with equation  $\frac{2x^2}{9} + \frac{(y-1)^2}{4} = 2$ .

**TRICK**

Write the equation in standard form and look at the graph.



## 8 – Coordinate geometry and conic sections (cont'd)

- 5 For the hyperbola with equation  $\frac{(x-a)^2}{4} - \frac{(y-4)^2}{16} = 1$ , one asymptote has equation  $y = -2x$ .

- (A) Find  $a$  and hence the equation of the hyperbola.  
(B) Sketch the graph of the hyperbola showing the coordinates of its centre and vertices and the asymptotes.

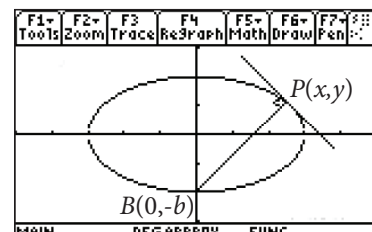
### HINT

Make sure that the graph's asymptotes cross at the centre and pass through the correct intercepts on the axes. Show the asymptotic behaviour clearly.

- 6 Points  $B(0, -b)$  and  $P(x_0, y_0)$  lie on the ellipse with equation  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ . The chord  $PB$  is normal to the ellipse at  $P$ .

- (A) The equation of the chord  $PB$  is  $a^2y_0x - b^2x_0y = (a^2 - b^2)x_0y_0$ . (**Do not derive or prove this.**) Use the fact that the chord passes through the point  $B(0, -b)$  to show that  $y_0 = \frac{b^3}{a^2 - b^2}$ ,  $b$  and  $-b$ .

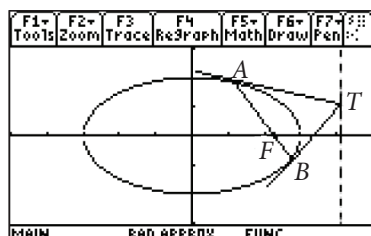
- (B) Hence find possible values of the eccentricity of the ellipse if  $y_0 = \frac{b^3}{a^2 - b^2}$ .



### HINT

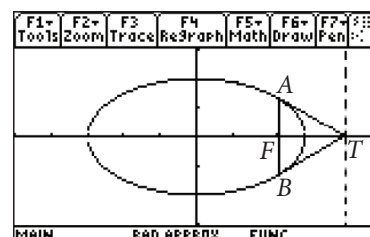
- ☞ If an equation is given then don't waste time re-deriving or proving it!
- ☞ Note that there are two distinct cases,  $x_0 = 0$  and  $x_0 \neq 0$ .
- ☞ Also, don't omit the case where  $y_0 \leq b$ .

- 7 Points  $A$  and  $B$  lie on the ellipse with equation  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ . The chord  $AB$  passes through the focus  $F(ae, 0)$  and the tangents to the ellipse at points  $A$  and  $B$  meet at point  $T(x_0, y_0)$ .



- (A) The equation of the chord  $AB$  is  $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$ . (**Do not derive or prove this.**) Use the fact that the chord passes through the focus to show that  $x_0 = \frac{a}{e}$ .

- (B) Hence state where the point  $T$  lies.  
(C) Find the point  $A$  such that the tangents to the ellipse at points  $A$  and  $B$  meet on the  $x$ -axis.  
(D) By finding  $AF$  and  $FT$ , find an expression for  $\angle ATF$ .  
(E) Find the area of  $\triangle ABT$ .



# 9 – Geometry in a circle

In any Geometry question you should be sure to draw your own diagram and mark the angles on it, not just on the question paper that you may not hand in. For success in Geometry questions you need to learn the basic terminology and provide the examiner with convincing reasoning.

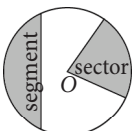


## THINGS TO KNOW

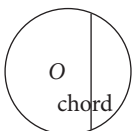
### Circle definitions and theorems:

- Supplementary angles sum to  $180^\circ$  and complementary angles sum to  $90^\circ$ .

- Sector and segment:

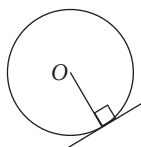


- A chord is a line joining two points on the circumference of a circle.

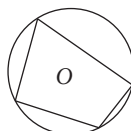


- Angle subtended at a point by a line: the angle is standing on the line or the line 'stretches under' the angle.

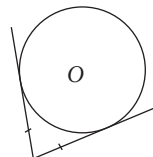
- A tangent is a line that touches a circle at only one point. A tangent to a point on a circle is perpendicular to the radius at that point.



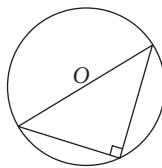
- A cyclic quadrilateral is a four-sided figure in a circle, with each vertex (corner) of the quadrilateral touching the circumference of the circle. The circle is the quadrilateral's circumcircle. The opposite angles of a cyclic quadrilateral add up to  $180^\circ$  (are supplementary).



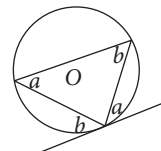
- Tangents to a circle from an external point are equal.



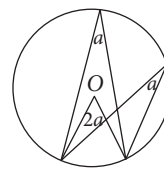
- Angles at the circumference subtended by a diameter are right angles.



- A tangent meets a chord at an angle equal to that subtended at the circumference by the chord (or the angle between the chord and the tangent is equal to the angle in the alternate segment).

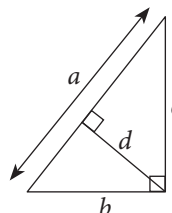


- Angles in a circle standing on the same arc (or subtended by the same chord) are equal.



- Angles at the centre are twice the angle at the circumference standing on the same arc (subtended by the same chord).

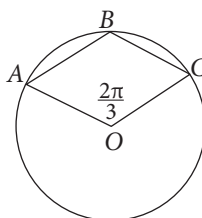
- By considering areas or otherwise, show that in the diagram given at the right,  $b^2c^2 = d^2(b^2 + c^2)$ .



### TRICK

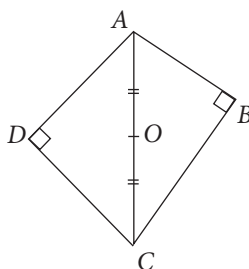
Similar triangles and Pythagoras' Theorem may help.

- A quadrilateral  $OABC$  is drawn in the circle centre  $O$  as shown. If  $\angle AOC = \frac{2\pi}{3}$ , show that  $OABC$  is a rhombus.



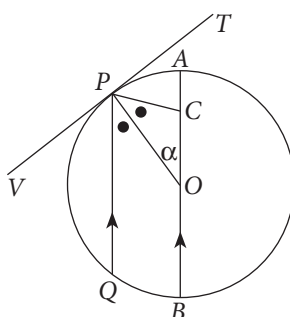
- Prove that  $ABCD$  shown in the diagram is a cyclic quadrilateral.

- If  $O$  is the midpoint of  $AC$ , prove that  $O$  is the centre of the quadrilateral's circumcircle.



- Hence show that  $AO + OB + OC + OD = 2AC$ .

- The diagram shows a circle centre  $O$  with tangent  $TV$  touching at point  $P$ . Chord  $PQ$  is parallel to diameter  $AB$ . Radius  $OP$  bisects  $\angle QPC$  and  $\angle POC = \alpha$  where  $0 < \alpha < \frac{\pi}{4}$ . Prove that  $\angle PCA = 2\alpha$ .



### TRAP

Be careful of the distinction between obtuse and reflex angles at the centre of a circle and don't mix up this quadrilateral with cyclic quadrilaterals.

### TRAP

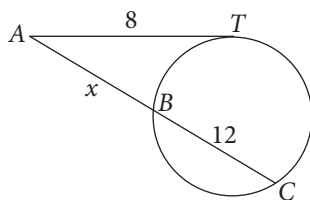
Give good, clear reasons in the proofs.

## THINGS TO KNOW

For a proof, you must show your geometrical argument clearly.



- 5 In the diagram the line  $AT$  of length 8 units is a tangent to the circle at  $T$  and  $AC$  is a secant meeting the circle at  $B$  and  $C$ . If chord  $BC$  has length 4 units, find the length of  $AB$ ,  $x$  units.



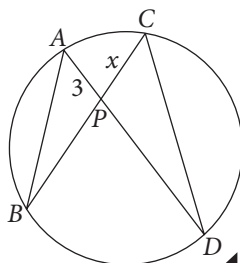
## THINGS TO KNOW

### Tangent-Secant Theorem:

If a tangent from an external point  $A$  meets the circle at  $T$  and a secant from the external point  $A$  meets the circle at  $B$  and  $C$  respectively, then  $AT^2 = AB \cdot AC$ .

Check the feasibility of your solution.

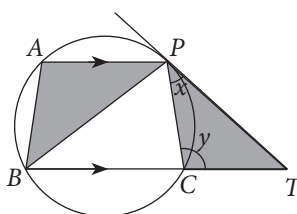
- 6 Two chords  $AD$  and  $BC$  to a circle intersect at point  $P$ .



- A Use similar triangles to show that  $\frac{AP}{CP} = \frac{BP}{DP} = \frac{AB}{CD}$

- B If  $AP$  has length 3 units,  $AD$  has length 8 units,  $CP$  has length  $x$  units and  $BC$  has length  $y$  units, show that  $x^2 - xy + 15 = 0$ .

- 7 In the diagram,  $TP$  is tangent to the circle and lines  $BT$  and  $AP$  are parallel.  $\angle TPC = x^\circ$  and  $\angle PCT = y^\circ$ .



- A Prove that the shaded triangles are similar.

- B If  $AB = CT$ , show that:

i  $AP = PC$

ii  $2x + y = \pi$ .

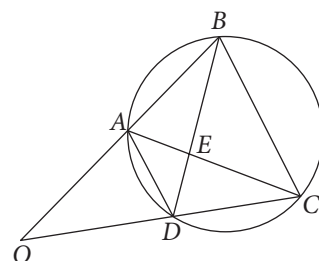
## TRAP

Be careful in the naming of the line segments and intervals and give complete explanations.

## HINT

You need to know the circle theorems and remember that proof of similarity means giving all the criteria, not just 2 angles. When you 'show' you must give reasons for similarity or congruence. Look for isosceles triangles in the last part.

- 8 The points  $A$ ,  $B$ ,  $C$  and  $D$  are placed on a circle of radius  $r$  such that chords  $BD$  and  $AC$  meet at  $E$ .  $BA$  and  $CD$  are produced (extended) to meet at point  $O$  and  $OAED$  is a cyclic quadrilateral.



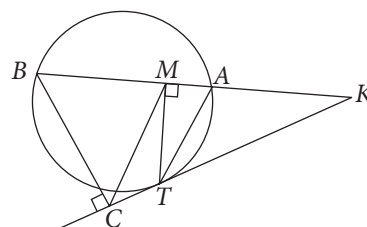
- A Find the size of  $\angle BDO$ , giving reasons for your answer.

- B Show that the length of  $BC$  is given by  $2r$ .

## HINT

You should redraw the diagram on your answer sheet so that the examiner can see which angles you are talking about. Make sure that you name the angles and lengths correctly. Don't assume anything unless you are given it or until you have proven it.

- 9 The diagram shows points  $A$ ,  $B$  and  $T$  marked on a circle. A tangent to the circle at  $T$ ,  $KC$ , is drawn such that  $BC$  is perpendicular to  $KC$ .  $TM$  is perpendicular to  $AB$ .



- A Show that  $BMTC$  is a cyclic quadrilateral.

- B Show that  $\angle MCT = \angle MBT$ .

- C Hence show that  $MC$  is parallel to  $AT$ .

## TRICK

Some people do not like the look of geometry questions and leave them out! If you do that then you get no marks but if you have a try by writing down what you know, you give yourself a chance to get some marks.

# 10 – Vector algebra and geometry

Different examinations provide different formula sheets and some have definitions in Vector Algebra. It is best if you still learn them off by heart and the best way to do this is to practise. You also need to be able to use the Algebra in Coordinate Geometry contexts.



## THINGS TO KNOW

If  $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$  and  $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$  then  $\vec{a} = \vec{b}$  if  $a_1 = b_1, a_2 = b_2$  and  $a_3 = b_3$ .

$\vec{a} + \vec{b} = ((a_1 + b_1)\vec{i} + (a_2 + b_2)\vec{j} + (a_3 + b_3)\vec{k})$  and unit vector  $\hat{a} = \frac{1}{|\vec{a}|}$ . Also,  $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_1b_1 + a_2b_2 + a_3b_3$ .

For a point with position vector  $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$  the distance from the origin is given by  $|\vec{r}| = \sqrt{x^2 + y^2 + z^2} = r$  and the unit vector is  $\hat{r} = \frac{x}{|\vec{r}|}\vec{i} + \frac{y}{|\vec{r}|}\vec{j} + \frac{z}{|\vec{r}|}\vec{k}$  where  $\frac{x}{|\vec{r}|}$ , etc. can be called direction cosines.

① True or false?  $\vec{a} \cdot \vec{b} \cdot \vec{c} = abc \cos\theta$ .

②  $\vec{a} = 2\vec{i} - p\vec{j} + 2\vec{k}$  and  $\vec{b} = 2\vec{i} + q\vec{j} - 4\vec{k}$  where  $p$  and  $q$  are real numbers. If  $|\vec{a}| = 3$  and  $3\vec{a} - 2\vec{b} = 2\vec{i} - 11\vec{j} + 14\vec{k}$

Ⓐ find  $p$  and  $q$

Ⓑ hence find  $\vec{a} \cdot (3\vec{a} - 2\vec{b})$ .

③ If  $\vec{OA} = 4\vec{i} - 2\vec{j} + \vec{k}$  and  $\vec{OB} = 5\vec{i} + 2\vec{j} + \vec{k}$ , find  $\vec{AB}$ .

④ On an Argand diagram,  $\vec{i}$  is a unit vector in the direction of the real axis and  $\vec{j}$  is a unit vector in the direction of the imaginary axis. If  $A = 3 + 0i$ ,  $B = -1 - 2i$  and  $C = 1 - 2i$ , express  $\vec{AB}$  in terms of  $\vec{i}$  and  $\vec{j}$ .

⑤ Ⓐ Prove that for any vectors  $\vec{a} = \vec{a}$  and  $\vec{b} = \vec{b}$ ,

$$\frac{\vec{a} \cdot (|\vec{b}|\vec{a} + |\vec{a}|\vec{b})}{|\vec{a}|} = |\vec{a}||\vec{b}| + \vec{a} \cdot \vec{b}.$$

Ⓑ Hence show that if  $\vec{c} = |\vec{b}|\vec{a} + |\vec{a}|\vec{b}$  where  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  are non-zero vectors as shown,  $\vec{c}$  bisects the angle between  $\vec{a}$  and  $\vec{b}$ .

Ⓒ For vectors  $\vec{a}[2,1,-2]$  and  $\vec{b}[4,0,3]$  (or using alternative notation,  $\vec{a} = 2\vec{i} + \vec{j} - 2\vec{k}$  and  $\vec{b} = 4\vec{i} + 3\vec{k}$ ), use your answer to part b to find the vector  $\vec{c}$  that bisects the angle between  $\vec{a}$  and  $\vec{b}$ .

⑥ Vector  $\vec{PQ}$  is such that the position vectors of  $P$  and  $Q$  are  $\vec{p} = 3\vec{i} + 4\vec{j} - 5\vec{k}$  and  $\vec{q} = a\vec{i} + b\vec{j} + 4\vec{k}$  respectively. If  $\vec{OP}$  is perpendicular to  $\vec{OQ}$ , find the relationship between  $a$  and  $b$ .

## TRICK

Remember, answers must be real numbers.

## HINT

Perhaps you should draw a diagram to work out whether to add or subtract vectors.

## TRAP

Watch the signs and use the correct vector notation. A diagram will help.

## HINT

Remember what is meant by  $\vec{a} \cdot \vec{a}$  and notice the words 'hence' and 'use your answer'.

## HINT

What do you know about the product of two vectors that are perpendicular?



- 7 Four points  $P, Q, R, S$  have position vectors  $\underline{p} = -2\hat{i} - 5\hat{j}$ ,  $\underline{q} = 2\hat{i} - 7\hat{j}$ ,  $\underline{r} = 6\hat{i} - 5\hat{j}$  and  $\underline{s} = 2\hat{i} + 3\hat{j}$  respectively with respect to origin  $O$ .

- (A) Find  $\vec{PR}$  and  $\vec{QS}$ .  
 (B) Hence show that  $\vec{PR}$  and  $\vec{QS}$  are perpendicular.

## HINT

- ☞ Drawing a diagram should help.
- ☞ You must say **why** the vectors are perpendicular.
- ☞ Remember  $\vec{PR} \neq \underline{p} - \underline{r}$  and the fact that  $\vec{PR} = \vec{PO} + \vec{OR}$  is of no help!

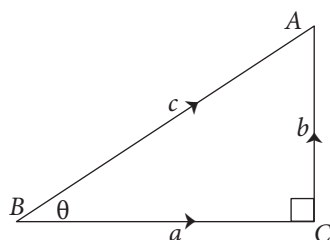
- 8  $\vec{OA} = 7\hat{i}$ , and  $\vec{OB} = 5\hat{i} + 4\hat{j}$  and  $\vec{OC} = -3\hat{i}$  lie on a circle.

- (A) Use a scalar product to show that  $\vec{AC}$  is a diameter of the circle.  
 (B) Find the centre of the circle using the properties of vectors.

## TRICK

Think about the properties of diameters and chords. Draw a diagram to help.

- 9  $ABC$  is a right angled triangle with sides  $\underline{a}$ ,  $\underline{b}$  and  $\underline{c}$ .

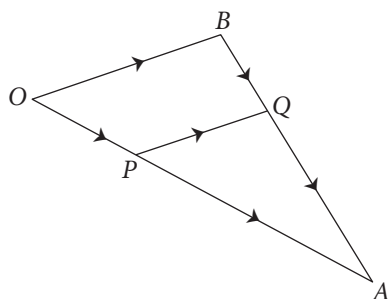


- (A) Show that  $\underline{c} \cdot \underline{b} = |\underline{c}||\underline{b}|\sin \theta$ .  
 (B) Prove that  $c^2 = a^2 + b^2$ .

## TRICK

Look for complementary angles in the first part and the meaning of  $c^2$  in the second part.

- 10 In the diagram,  $\vec{OA} = \underline{a}$ ,  $\vec{OB} = \underline{b}$ ,  $\vec{OP} = \frac{1}{2}\vec{PA}$  and  $\vec{BQ} = \frac{1}{2}\vec{QA}$ .



- (A) Find each of the following in terms of  $\underline{a}$  and  $\underline{b}$ :

- (i)  $\vec{AB}$   
 (ii)  $\vec{OP}$

- (B) Use a vector proof to show that  $\vec{PQ} \parallel \vec{OB}$ .

- (C) Find the area of  $\triangle OAB$  and  $\triangle PAQ$  in terms of  $a$  and  $b$  if  $\angle AOB = 60^\circ$ .

# 11 – Differentiation

The idea that differentiation is related to the gradient of a curve leads to some standard results. Different examinations provide different formula sheets but you should still learn your list off by heart. The best way to do this is to practise.



- ① Many of these standard derivatives will be on your list to learn. Cross out the ones you don't need and write down in your own notation the ones you need to learn:

## THINGS TO KNOW

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(\log_e x) = \frac{1}{x}$$

$$\frac{d}{dx}(\sin ax) = a \cos ax$$

**Note:**  $\ln x = \log_e x$ ;  $x > 0$

$$\frac{d}{dx}(\tan ax) = a \sec^2 ax$$

$$\frac{d}{dx}(\cos ax) = -a \sin ax$$

$$\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}; a \neq 0$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}); x > a > 0$$

$$\int \frac{-1}{\sqrt{a^2 - x^2}} dx = \cos^{-1} \frac{x}{a}; a > 0, -a < x < a$$

**Chain rule:**

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

**Quotient rule:**

**Product rule:**

$$\frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{d\left(\frac{u}{v}\right)}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

- ② Differentiate  $y = x^2(3 - x)$ .

**TRAP**

You don't need the product rule, just multiply it out first!

- ③ Differentiate  $(1 + x)^n - 1$ .

**TRAP**

Watch the  $-1$ .

- ④ A Find  $\frac{d}{dx}(\sqrt{2x})$ .

- B Hence differentiate  $y = \tan^{-1}(\sqrt{2x})$ .

**TRAP**

$\sqrt{2x} \neq \sqrt{2}x$  and  $\sqrt{2}x = x\sqrt{2}$ .  
Remember to use the 'hence'.



5 Find:

A  $\frac{d}{dx} \sin^{-1}(2x^2)$

B  $\frac{d}{dx} (3 \cos^{-1} 2x)$

**TRAP**

You don't need the product rule!

**HINT**

☞ You need to know your standard derivatives and how to use the chain rule.

☞ Be careful when you are dealing with the  $2x^2$ , when handling negatives and  $(2x^2)^2$ , and remember that  $\sin^{-1} x \neq \sin(x)^{-1}$ 6 A Show that  $\sqrt{2} \sin(x + \frac{\pi}{4}) = \sin x + \cos x$ B If  $y = e^x \sin x$ :

I show that  $\frac{dy}{dx} = e^x (\sin x + \cos x)$

II and hence show that  $\frac{dy}{dx} = \sqrt{2} e^x \sin(x + \frac{\pi}{4})$

III find  $\frac{d^2y}{dx^2}$ .

**HINT**

Use the relationship in part a to help you with part b iii.

7 State the domain for each of the following rules for differentiation:

A  $\frac{d(\sin^{-1} x)}{dx} = \frac{1}{\sqrt{1-x^2}}$

B  $\frac{d(\sin^{-1} \frac{x}{a})}{dx} = \frac{1}{\sqrt{a^2-x^2}}$

**TRAP**

Read the question carefully and notice the principal values.

8 If  $x = a \cos t$  and  $y = a \sin t$  where  $a \neq 0$  and  $t$  is a parameter, show that  $\frac{dy}{dx} = -\cot t$ .9 If  $f(x) = (ax + b)(3 - 2x)^{\frac{3}{2}}$  where  $a$  and  $b$  are integers then  $f'(x) = 5x(3 - 2x)^{\frac{1}{2}}$ . Use differentiation to find the values of  $a$  and  $b$ .**TRICK**

☞ Choose the correct method of differentiation and watch out for the chain rule.

☞ Getting  $f'(x)$  in the right form to equate coefficients will help.10 The polynomial  $P(x)$  satisfies the equations  $P(1) = 0$  and  $P'(1) = 0$ .A Show that  $(x - 1)^2$  is a factor of  $P(x)$ .B Hence show that for  $P(x) = x^4 - 2x^3 + 2x^2 - 2x + 1$ ,  $P(1) = 0$  and  $P'(1) = 0$  and that  $x = 1$  is the only real zero of  $P(x)$ .**HINT**

☞ Make use of the Factor theorem and use the product rule to continue the proof.

☞ Note the fact that  $(x + 1)^2$  is a factor and that there are no other factors of the form  $(x - a)$  where  $a$  is real means that 'there is only one zero'.

# 12 – Gradients, tangents and normals

Not only will you need to know how to differentiate various functions but you also need to understand how the derivative relates to the gradient of the tangent to the curve of the function and to be able to find the equation of this tangent at a given point.



## THINGS TO KNOW

The gradient of the tangent to  $y = f(x)$  at the point  $(x_1, y_1)$  is given by  $f'(x_1)$  and the gradient of the normal by  $\frac{-1}{f'(x_1)}$ .

- ① Find  $f'(2)$  if  $f(x) = 2x^3 - x$ .
- ② For the curve with equation  $y = 3e^x + 2x$ , find the value of  $x$  for which the gradient of the tangent is 5.
- ③ For the curve with equation  $y = 2\log_e x$ , find:
  - A the gradient of the tangent to the curve at the point  $(e, 2)$  and hence
  - B the equation of the tangent to the curve at the point  $(e, 2)$ .
- ④ For  $f: \mathbb{R} \rightarrow \mathbb{R}$  where  $f(x) = x^3(x - 3)$ , find the equation of the tangent at the point where  $x = 2$ .

## THINGS TO KNOW

Know the method!  $f'(x)$  gives the gradient of the tangent to the curve with equation  $y = f(x)$  at any point  $x$ . The equation of the tangent to this curve at the point  $(x_1, y_1)$  can be found by using the formula  $\frac{y - y_1}{x - x_1} = m$  where  $m = f'(x)$ .

## TRAP

Read the question! Yes, it says 'find the equation' so get the equation, not just the gradient. It's probably easier to multiply out the brackets before differentiating.

- ⑤ For what value/s of  $k$  is the line with equation  $y = 3x + k$  a tangent to the curve with equation  $y = x^3$ ?
- ⑥ Use the gradient of a chord to find the approximate angle to nearest degree that the curve with equation  $y = x(4 - x^2)^{\frac{1}{4}} + \frac{1}{(4 - x^2)^{\frac{1}{4}}}$ ;  $-2 < x < 2$  makes with the horizontal at the point where the graph cuts the  $x$ -axis at  $x \approx -0.5063$ .
- ⑦ If  $y = \frac{2}{1+x} \tan^{-1}(x) + \frac{1}{1+x} \log_e(1+x^2)$ :

## HINT

You should know the standard equation of a line  $y = mx + c$  and how to get the gradient of a curve. Always look for the easiest way to solve a problem. There may be more than one value of  $k$ .

## TRICK

Use trigonometry on two points near the intercept as you will need to use an analytical rather than a calculus method.

- A show that  $\frac{dy}{dx} = \frac{2}{1+x^2} - \frac{1}{(1+x)^2} \log_e(1+x^2) - \frac{2}{(1+x)^2} \tan^{-1}(x)$ .

## HINT

Use the product rule (easier than the quotient rule perhaps) and be careful when dealing with the log of a function. Combining the terms in log and  $\tan^{-1}$  is perhaps the hardest part so be careful not to get lost in the Algebra. Remember that  $\tan^{-1}$  means the principal value.

- B Hence verify that  $y$  satisfies the differential equation  $\frac{dy}{dx} + \frac{y}{1+x} = \frac{2}{1+x^2}$ .

## TRAP

When substituting the given information in the last part, watch out for arithmetic errors.

- C Find  $y$ , correct to three decimal places when  $x = 0.75$ .

## HINT

Remember, you need radian mode.

# 13 – Sketch graphs and calculus

Sketch does not mean a rough or careless attempt. You are well advised to use pencil so that the result is not messy with multiple lines making it hard to read. Make sure that you have a reasonable scale, the correct domain, the asymptotes and intercepts with the axes clearly shown and labelled. Also make the asymptotic behaviour very clear. Mark end points if appropriate.



1 Complete the following table for a graph of  $y = f(x)$  with words and curves:

|                                                         | $f'(x) < 0$<br>Gradient:<br>negative | $f'(x) = 0$<br>Gradient: | $f'(x) > 0$<br>Gradient: |
|---------------------------------------------------------|--------------------------------------|--------------------------|--------------------------|
| $f''(x) < 0$<br>Rate of change of gradient:<br>negative |                                      |                          |                          |
| $f''(x) = 0$                                            |                                      |                          |                          |
| $f''(x) > 0$                                            |                                      |                          |                          |

2 For the function with rule  $f(x) = (x - a)^4$  where  $a$  is a positive constant, find:

- A  $f'(a)$
- B  $f''(a)$ .

Hence describe the curve at  $x = a$ .

TRAP

Perhaps you need to use extra information to describe the curve.

3 For the function  $f(x) = ax^3 - 3ax + 1$  where  $a > 0$ :

- A show that  $f(x)$  has stationary points at  $x = \pm 1$ .
- B show that  $f(x)$  has exactly one zero when  $a < \frac{1}{2}$ .

TRICK

When you are asked to 'show' something it is a good idea to note your intentions as well as giving your reasoning. Graphs will help you with the explanation. Watch your Algebra and make sure your argument is valid.

4 If  $f(x) = \tan x + 2\cos x$  where  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ :

- A show that  $f'(x) = \frac{2\sin^3 x - 2\sin x + 1}{\cos^2 x}$  where  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
- B show that the graph of  $f(x)$  has no stationary points
- C deduce that  $f(x)$  has an inverse function.

HINT

Drawing the graph will help. Also remember the properties of functions!



- 5 For the function  $f(x) = \sin^{-1}(x+3)$ ,
- A find the gradient of the curve at the point where  $x = -3$
- B sketch the graph of  $y = f(x)$  for  $-4 \leq x \leq -2$ .

**SLIP**

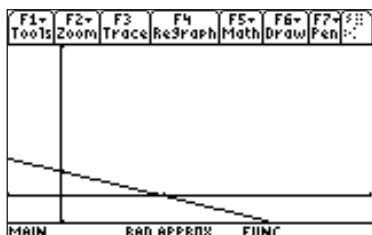
Make sure that your graph, axes and endpoints are clear (what is the domain?) and make sure that the curve is the right way up.

**HINT**

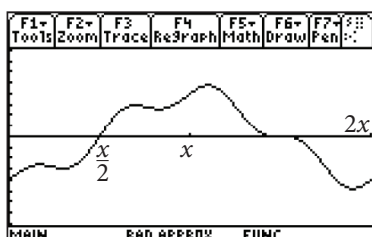
There is no need to expand  $(x+3)^2$  because you get complicated algebra.

Answers of  $\frac{1}{\sqrt{1-x^2}}$  and  $\frac{1}{\sqrt{1+(x+3)^2}}$  for the gradient are wrong!

- 6 For the function  $f(\alpha) = 2 + \sqrt{3} \sin \alpha - \cos \alpha$ :
- A find  $f'(\alpha)$
- B find exact values of  $\alpha$  such that  $f'(\alpha) = 0$ ,  $0 \leq \alpha \leq 2\pi$
- C sketch the graph of  $y = f(\alpha)$  for  $0 \leq \alpha \leq 2\pi$ .
- 7 The graph of  $y = f(x)$  is shown. On the same axes, draw the graph of  $y = F(x)$ , the antiderivative of  $f$ .



- 8 The graph of  $y = f(x)$  is shown. If  $F(x)$  is the antiderivative of  $f$ , state the values of  $x$  of the stationary points of the graph of  $y = F(x)$ .



- 9  $m(t) = e^{\cos 2\pi t}$  and  $n(t) = \cos(2\pi(t+0.5)) + 1$ .
- A Draw a rough sketch of the graphs of  $m(t)$  and  $n(t)$  on the same axes for  $0 \leq t \leq 2$ .
- B On the same axes as part a, draw a rough sketch of the graph of  $y(t) = m(t) + n(t)$ .
- C Find the period of each of  $m(t)$  and  $n(t)$ .
- D Find the maximum and minimum values of  $m(t)$  and  $n(t)$  for  $0 \leq t \leq 2$ .
- E Hence describe a relationship between the maximum and minimum values of  $m(t)$  and  $n(t)$ .

**SLIP**

Be careful of sloppy graph shapes. Remember to show all scales, label the axes and mark relevant points. Your calculator must be in radian mode.

# 14 – Maximum and minimum of functions

Usually a maximum or minimum question requires Calculus techniques and if you sketch the graph or have an idea of what the graph looks like then you may be able to see the answer without Calculus. Either way, a graph shows you if your answer is sensible. Remember that a graph from a calculator is likely to be acceptable if the question is worth only 1 mark.



① For the function  $f(x) = e^{1-x} \sin 2x$ :

(A) find  $f'(x)$

(B) show that there are stationary points when  $\tan 2x = 2$

(C) find the values of  $x$  for which there are maximum and minimum values of the function.

## THINGS TO KNOW

The general solution of trigonometric equations:

$$\sin \theta = a \text{ then } \theta = n\pi + (-1)^n \sin^{-1} a$$

$$\cos \theta = a \text{ then } \theta = 2n\pi \pm \cos^{-1} a$$

$$\tan \theta = a \text{ then } \theta = n\pi + \tan^{-1} a$$

Note the capital letters on the inverse trigonometric functions mean the principal value or smallest possible value.

② For the function  $f(x) = \frac{20}{1 + 19e^{-x}}$ ,

(A) sketch the graph of  $f$  for  $-3 \leq x \leq 3$  and state the equation of the horizontal asymptote

(B) on the graph, mark the point,  $P$ , that corresponds to the maximum rate of change of  $f$  with  $x$ .

## TRAP

Maximum rate of change means maximum gradient.

③ (A) For the function  $f(x) = \frac{x+h+k}{3(hkx)^{\frac{1}{3}}}$  where  $x > 0$  and  $h$  and  $k$  are positive real numbers, find the value of  $x$  for which the function is a minimum.

(B) Assuming that  $\frac{h+k}{2} \geq \sqrt{hk}$ , show that for  $g$  a positive real number,

$$\left( \frac{g+h+k}{3\sqrt[3]{ghk}} \right)^3 \geq \left( \frac{h+k}{2\sqrt{hk}} \right)^2.$$

(C) Hence deduce that  $\frac{g+h+k}{3} \geq \sqrt[3]{ghk}$ .

## TRAP

Differentiate carefully and watch for the chain rule on the denominator. Alternatively you might find it easier to divide each term of the numerator by  $3(hkx)^{\frac{1}{3}}$ . You must check that you have a minimum, either from  $f''(x) > 0$  or by looking at  $f'(x)$  on either side of the minimum point.

## TRICK

It will be useful to write  $f'(x)$  in the form

$$\frac{2x-h-k}{9\sqrt[3]{hk}x^{\frac{4}{3}}} \text{ to help to convince the examiners of the 'show'.$$



- 4 The function  $f(x) = 6ab + \frac{3}{2}a^2\left(\frac{\sqrt{3} - \cos x}{\sin x}\right)$  where  $0 < x < \frac{\pi}{2}$  and  $a$  and  $b$  are constants.

- A Find  $f'(x)$ .  
 B Hence find the value of  $x$  correct to 3 decimal places such that  $f(x)$  is minimised.  
 C Show that the minimum value of  $f(x)$  is  $6ab + \frac{3}{\sqrt{2}}a^2$ .

## TRAP

Be careful since the value of  $x$  is not exact; it is not a 'special angle'. Also you must justify the minimum using a sign diagram or the second derivative.

## TRAP

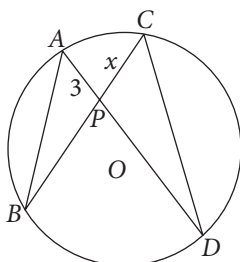
The answer given is exact so you can't use the approximate value of  $x$  you got before; use your knowledge of trigonometric functions instead.

- 5 The maximum and minimum values of  $y = e^{1-x} \sin 2x$  for  $0 \leq x \leq \pi$  occur when  $x = \frac{\phi}{2} + \frac{k\pi}{2}$  for  $k = 0, 1$  and  $\tan \phi = 2$ . Show that the ratio of the maximum value to the minimum value of  $y$  is  $e^{\frac{\pi}{2}}$ .

## TRICK

Keep brackets on where possible, make sure that you write the ratio correctly and remember that  $e^{-x} \neq -e^x$ .

- 6 Two chords  $AD$  and  $BC$  to a circle centre  $O$  intersect at point  $P$ . When  $AP$  has length 3 units,  $AD$  has length 8 units,  $CP$  has length  $x$  units and  $BC$  has length  $y$  units, it can be shown that  $x^2 - xy + 15 = 0$ . Find the length of the shortest chord that passes through  $P$ .



## HINT

$y$  is a variable and you need to differentiate with respect to  $x$ . This could be done implicitly but it is probably easier to make  $y$  the subject first. An elegant method involves the discriminant  $\Delta = y^2 - 60 \geq 0$  for  $x$  real.

- 7 A machine moves up and down according to the rule  $y = 30 + 25 \sin\left(\frac{(4t-1)\pi}{2}\right)$ ;  $t \geq 0$ . The distance of the machine from the ground is  $y$  metres at time  $t$  minutes from the start.
- A What is the maximum value of  $y$ ?  
 B What is the minimum value of  $y$ ?  
 C After how long from the start is the first minimum?  
 D After how long does the machine reach a height of 40 metres from the ground? Give your answer correct to 4 decimal places.  
 E For how long during the first minute is the machine more than 40 metres from the ground?

## HINT

Make sure you are in radians and that your answers do not have  $\pi$  in them. The table on your calculator will probably give whole numbers, remember. Did you use calculus? You don't have to!

# 15 – Rates of change

The rate of change of one variable with respect to another can be thought of as a measure of the gradient of the graph of their relationship. Alternatively, questions can be asked that require you to set up a differential equation. Rates questions are easy to identify because the word 'rate' is usually in them somewhere! In worded problems you must use the correct units and put the correct sign (+ or –) on the rate.



## THINGS TO KNOW

When no 'with respect to' is mentioned, the rate of change of a variable is a measure of the gradient of the graph of the relationship with time. So for  $y = f(x)$ , the rate of change of  $y$  with respect to  $x$  is  $\frac{dy}{dx}$  and the rate of change of  $y$  is  $\frac{dy}{dt}$ .

The average rate of change between two points is the measure of gradient of a secant or chord of the graph between those points.

- ① Find the average rate of change of the curve with equation  $y = x^2 + 3x$  between  $x = 3$  and  $x = 3.01$ . Compare this with the exact rate of change of the curve  $y = x^2 + 3x$  at  $x = 3$ .
- ② The needle on a sewing machine moves up and down according to the rule  $y = 25 + 20 \sin\left(\frac{(2t-1)\pi}{2}\right)$ ;  $t \geq 0$  where  $y$  mm is the distance of the needle from the base at time  $t$  seconds from the start.
  - A Find the rate of change of  $y$  with respect to  $t$  in simplest form.
  - B Find the value of the rate of change of  $y$  with respect to  $t$  when  $t = 1$ .
  - C Find the value of rate of change of  $y$  with respect to  $t$  when  $t = \frac{1}{2}$  correct to 1 decimal place.

## TRAP

A common error was to omit pi in expressions or give an answer with pi in it when asked for a numerical answer.

## THINGS TO KNOW

Rates can be related; when one variable changes, other variables that are related to it will also change. The chain rule will be useful when rates are related. So, for example, if  $y = f(x)$  and  $x = g(t)$ , then  $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$  will give the rate of change of  $y$  with respect to  $t$  in terms of the other two rates. Be careful to read the question to make sure that what looks like a rate question actually involves a rate.

- ③  $y = 100x^{\frac{3}{2}}$  and  $\frac{dy}{dt} = 4$ .
  - A Find  $\frac{dy}{dx}$ .
  - B Write a chain rule that could be used to find  $\frac{dx}{dt}$ .
  - C Hence find an expression for  $\frac{dx}{dt}$ .
  - D Find the exact value of the rate of change of  $x$  when  $x = 16$ .

## TRICK

The chain rule doesn't necessarily have the derivatives the right way up!

You may need to use the fact that  $\frac{dy}{dx} = 1 \div \frac{dx}{dy}$ .

- ④ If the area of a circle is increasing at a rate of  $20 \text{ cm}^2$  per second, at what rate is the radius increasing when the radius is  $1 \text{ cm}$ ?

## TRICK

Draw a labelled diagram.

- ⑤ The cost  $C$  dollars of producing  $v$  televisions per week is given by  $8000 + 75v - 0.02v^2$ . The revenue  $R$  dollars received from the sale of the televisions is  $135v - 0.01v^2$ .

- A Show that the profit  $P$  dollars when  $v$  televisions are made and sold is given by  $-8000 + 60v + 0.01v^2$ .

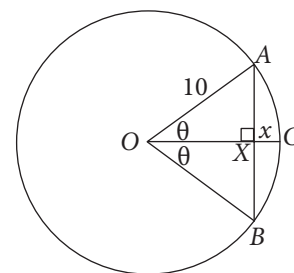
- B If the rate of production is being reduced at the rate of 15 televisions per week, at what rate will the profit change when 600 televisions are produced per week?

## HINT

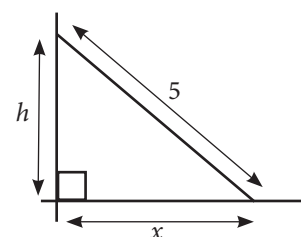
Notice that this question is a related rates question, but watch the signs.



- 6 A chord  $AB$  is drawn in a circle radius 10 cm centre  $O$ . The chord subtends an angle  $2\theta$  at the centre and radius  $OC$  meets  $AB$  perpendicularly at  $X$ . If  $XC$  is defined to be  $x$  as shown in the diagram:



- (A) show that  $\cos \theta = \frac{10 - x}{10}$
- (B) show that  $\frac{d\theta}{dt} = \frac{1}{10 \sin \theta} \frac{dx}{dt}$
- (C) explain why  $\frac{dx}{dt}$  is negative when the radius becomes smaller.
- 7 One end of a straight ladder of length 5 metres is sliding down a vertical wall, while the other end is sliding over horizontal ground. The whole motion occurs in a vertical plane. Let  $h$  metres be the height of the top of the ladder above the ground and  $x$  metres be the horizontal distance between the wall and the bottom of the ladder.



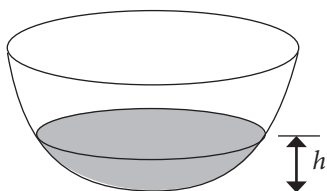
- (A) Find an expression for  $x$  in terms of  $h$ .
- (B) When the top end of the ladder is 4 metres from the ground, it is sliding down at  $0.5 \text{ ms}^{-1}$ . At what speed is the other end then moving over the ground?

### TRAP



Be careful in your application of Pythagoras' Theorem and with your differentiations and substitutions. Also think about the importance of + and - in the calculation of rates. You can use implicit differentiation if you wish.

- 8 A hemispherical bowl of radius  $R$  cm is being filled with liquid at a constant rate of  $k \text{ cm}^3$  per second. When the depth of water in the bowl is  $h$  cm, the volume,  $V \text{ cm}^3$ , of the water is given by  $V = \frac{\pi}{3}h^2(3R - h)$ . The bowl is initially empty.



- (A) Without proving the expression for the volume, show that  $\frac{dh}{dt} = \frac{k}{\pi h(2R - h)}$ .
- (B) Hence or otherwise find the time in seconds it takes to fill the bowl:
- (I) to a depth of  $\frac{1}{3}R$
- (II) to a depth of  $\frac{2}{3}R$
- (III) full to the brim.

### TRAP

Remember that  $R$  is a constant, not a variable. And don't prove the formula for  $V$ !

### HINT

Show all your working.

# 16 – Differential equations

Differential Equations can be first order  $\left(\frac{dy}{dx}\right)$  or second order  $\left(\frac{d^2y}{dx^2}\right)$ .

Remember that a general solution to a differential equation will have '+ C' whereas a particular solution can be found using given information or boundary conditions that allow you to find the value of the 'C'.



- ① Show that  $y = 7e^{-0.5t} + 2$  is a solution to the differential equation  $\frac{dy}{dt} = 1 - \frac{y}{2}$ .

**HINT**

Trying to solve the given differential equation is the harder way to do it and will result in errors. Stating a standard form of the solution is not 'showing'. Just differentiating and substituting is the easiest way.

- ② Show that the differential equation  $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4y = -4\sin(2x)$  is satisfied by  $y = \cos(2x)$ .

**SLIP**

Careful with the signs when you substitute in the LHS of the differential equation.

- ③ Solve the differential equations:

A  $\frac{dy}{dx} = \frac{1}{y}$ ;  $y \neq 0$  and  $x = 0$  when  $y = 2$ .

B  $\frac{dy}{dx} = \sqrt{1 - y^2}$  and  $y = 1$  when  $x = 0$ .

**HINT**

When you substitute the values for  $x$  and  $y$  to find + c, you need to decide whether to take the positive or negative square root. Look at the domain on which the given conditions sit and use that to decide.

- ④ Complete each of the following questions.

A Verify that  $x = 3 - \frac{1}{t+1}$  satisfies the differential equation  $\frac{dx}{dt} = (3-x)^2$ .

B If  $x = 3 - \frac{1}{t+1}$  then show that  $\frac{dx}{dt} = (3-x)^2$ .

C Solve the differential equation  $\frac{dx}{dt} = (3-x)^2$ .

D If  $\frac{dx}{dt} = (3-x)^2$  and  $x = 2$  when  $t = 0$ , find  $x$  as a function of  $t$ .

E If  $\frac{dx}{dt} = (3-x)^2$ , find  $\frac{d^2x}{dt^2}$  as a function of  $x$ .

**TRAP**

These all look like the same question but different ways of answering are required.

- ⑤ A Solve the differential equation:  $\frac{dA}{dt} = m - nA$ .

B If  $m = n = 1$  and  $A_0$  is the initial value of  $A$ , solve the differential equation and hence show that  $A = (1 - e^{-t}) + A_0e^{-t}$ .

C Hence find  $A$  correct to 2 decimal places when  $t = 0.5$  if initially  $A = 10$ .

**TRAP**

Careful how you deal with the initial conditions!

**HINT**

You can do this question even if you didn't get part b done! Don't waste the marks by leaving it out!



- ⑥ If  $y = \frac{2}{x+1} \tan^{-1}(x) + \frac{1}{x+1} \log_e(1+x^2)$ , show that  $\frac{dy}{dx} = \frac{2}{(1+x^2)} - \frac{2 \tan^{-1}(x)}{x+1} - \frac{\log_e(1+x^2)}{x+1}$ .

**TRAP**

Antidifferentiation will not be successful. You can't assume that  $y$  is a solution of the differential equation. Product rule is better than the quotient rule. Be careful of the chain rule needed with the  $\log_e(1+x^2)$  part. You will need to collect like terms in  $\log_e$  and  $\tan^{-1}$  to get the answer so don't get lost in the Algebra.

- ⑦ Given that  $\frac{dy}{dx} = \frac{1}{3x-1}$  where  $y = 0$  when  $x = 0$ , solve for  $y$  as a function of  $x$ .

**HINT**

Check the sign of  $\frac{1}{3x-1}$  with the given values of  $x$  and  $y$  before proceeding to antidifferentiate.

- ⑧ Solve the differential equation  $\frac{dy}{dx} = x\sqrt{x^2-25}$ ;  $x \geq 5$  given that  $y = 500$  when  $x = 13$ .

**TRAP**

This is not a differentiation/product rule question!

**HINT**

Using change of variable, you need to pick the substitution that makes the question easiest. Remember that you need to put in  $+c$  and be careful with your Algebra and evaluating the power  $\frac{3}{2}$ . Of course, correct notation is essential.

- ⑨ Find the value of  $k$ , an integer, for which  $y = \cos(2x)$  is a solution to the differential equation  $\frac{d^2y}{dx^2} + k\frac{dy}{dx} = \sin(2x)$ .

**SLIP**

Be careful of your Algebra and with equating coefficients.

- ⑩A If  $\frac{dv}{dt} = g - kv$  where  $g$  and  $k$  are constants, verify that  $v$  satisfies the differential equation  $\frac{d}{dt}(ve^{kt}) = ge^{kt}$ .

**TRICK**

A common error is to use the product rule incorrectly or try direct differentiation incorrectly. Notice 'hence' in the second question, so use the result from the first question.

- ⑩B Hence show that  $v = \frac{g}{k} - \left(\frac{g}{k} - v_0\right)e^{-kt}$  where  $v_0$  is the value of  $v$  when  $t = 0$ .

# 17 – Practical problems involving differential equations

Practical problems that can be written in the form of a differential equation are of many types. Some of these include situations such as salt or sugar in solution, heating and cooling, and exponential growth and decay. Setting up the differential equation is half the task, the other half being to solve it under the given conditions. Sometimes the DE can't be solved.



## THINGS TO KNOW

According to Newton's Law of Cooling, the rate of change of the temperature of an object  $\frac{dT}{dt} = -k(T - S)$  is proportional to the difference between its own temperature  $T^\circ\text{C}$  and the temperature of the surroundings,  $S$ .

- ① Newton's Law of Cooling states that the rate of cooling of a body is proportional to the excess of its temperature above that of its surroundings.
- Ⓐ If  $\frac{dT}{dt}$  is the rate of cooling in degrees Celsius per minute of a body at  $T^\circ\text{C}$  and  $\frac{dT}{dt} = -k(T - S)$ , state the temperature of the surroundings and explain the negative in front of the  $k$ .
- Ⓑ A kettle heated to a temperature of  $70^\circ\text{C}$  is placed on the bench where the temperature is maintained at  $20^\circ\text{C}$ . If, after 10 minutes, the temperature of the body is  $55^\circ\text{C}$ , find the temperature correct to one decimal place after a further 15 minutes.
- ② A thermometer measures the temperature of a cup of coffee in a room at  $25^\circ\text{C}$ . The cup of coffee is then moved outside and after 5 minutes the thermometer reads  $15^\circ\text{C}$ . After another 5 minutes it reads  $10^\circ\text{C}$ . Use Newton's Law of Cooling  $\frac{dT}{dt} = -k(T - S)$  to find the temperature outside.
- ③ Water in a saucepan at a temperature of  $T^\circ\text{C}$  loses heat to its surroundings at a rate of  $\frac{T - 20}{4}^\circ\text{C}$  per minute. When the saucepan is placed on the stove it can be heated up at  $25^\circ\text{C}$  per minute. The temperature of the water after time  $t$  minutes satisfies a differential equation. If initially the water in the saucepan is at  $20^\circ\text{C}$ , copy and complete the differential equation to find the initial temperature of the water:
- $$\frac{dT}{dt} = \underline{\hspace{1cm}}(T \underline{\hspace{1cm}}); t = 0, T = \underline{\hspace{1cm}}$$
- ④ A cup of hot water initially at a temperature of  $35^\circ\text{C}$  is placed in the refrigerator to cool it down. The temperature in the freezer is  $-1^\circ\text{C}$ . The rate at which the water cools is proportional to the difference between the temperature  $T$  of the water and the temperature of the refrigerator. So according to Newton's Law of Cooling,  $\frac{dT}{dt} = -k(T + 1)$  where  $t$  is the number of minutes after the cup was placed in the freezer.
- Ⓐ Show that  $T = Ae^{-kt} - 1$  satisfies the differential equation.
- Ⓑ Hence use the given conditions to find the value of  $A$ .
- Ⓒ If the temperature of the water after 5 minutes is  $17^\circ\text{C}$ , find the value of  $k$  correct to 4 decimal places and hence the expression connecting  $T$  and  $t$ .
- Ⓓ Find the temperature of the water in the cup after 10 minutes.

## HINT

Best to differentiate  $T$  expression rather than integrating the  $\frac{dT}{dt}$  expression. Be careful with the value of  $A$ . It is not 35.



## 17 – Practical problems involving differential equations (cont'd)

- 5 Falling into cold water is particularly dangerous because you lose body heat 25 times faster in cold water than in cold air. The normal body temperature is  $37^{\circ}\text{C}$ . Shivering begins at an approximate temperature of a  $36^{\circ}\text{C}$ , amnesia at  $34.4^{\circ}\text{C}$ , unconsciousness at  $30^{\circ}\text{C}$  and death at  $26^{\circ}\text{C}$ . The body temperature of a person,  $T^{\circ}\text{C}$ , who has been in the water for  $t$  minutes can be modelled by the differential equation  $\frac{dT}{dt} = k(T - T_w)$  where  $T_w$  is the temperature of the water and  $k$  is constant.
- A If the water temperature is  $2^{\circ}\text{C}$ , a person is expected to start shivering after 5 minutes and the expected time of survival is 80 minutes, solve the differential equation to show that the body temperature of a person at time  $t$  is given by  $T = 2 + 35e^{-0.0058t}$ .
- B Hence find for how long a person can stay in the water before becoming unconscious. Give your answer correct to 1 decimal place.
- 6 Sugar dissolves in a bowl of water at a rate equal to 4% of the amount of undissolved sugar. Initially the amount of undissolved sugar is 3 g and after  $t$  minutes  $x$  grams has dissolved. Write down the differential equation that models this process.
- 7 The number  $n$  of bacteria in a colony grows at a rate proportional to  $n^2$  according to the differential equation  $\frac{dn}{dt} = \frac{1}{1500}n^2$ ;  $0 \leq t < 15$  and  $n = 50$  when  $t = 0$ .
- A Solve the differential equation to show that  $n = \frac{1500}{15 - t}$ .
- B Explain briefly the domain restriction on  $t$ .
- 8 The amount of active ingredient in a medicine changes with time such that the rate of change of the amount,  $A(t)$  units, after time  $t$  days,  $t \geq 0$ , is given by  $\frac{dA}{dt} = 100 - A$ . If the medicine initially contains 20 units:
- A solve the differential equation to show that the amount of active ingredient  $A(t)$  units present after time  $t$  days is  $A(t) = 100 - 80e^{-t}$ .
- B Sketch the graph of  $A(t)$  against  $t$  for  $t \geq 0$ . Make sure you show clearly the limiting behaviour of  $A(t)$ . The amount of active ingredient in the medicine can be found by measuring the change in its colour  $C$  units and the rate of change of colour intensity  $C$  units with amount of active ingredient  $A$  can be found from the formula  $\frac{dC}{dA} = \frac{C(20 - C)}{20}$ .
- C Simplify  $\frac{1}{C} + \frac{1}{20 - C}$  by expressing with a common denominator.
- D Use the result from part c to show that  $A = \log_e \left( \frac{C}{20 - C} \right) + k$  where  $k$  is a constant.
- E Given that the medicine has 1 unit of colour, that is  $C = 1$ , when there is no active ingredient, that is  $A = 0$ , find the value of  $k$  and hence show that  $C = \left( \frac{20}{1 + 19e^{-A}} \right)$ .

**TRAP**

Notice the word 'undissolved'.

**TRAP**

'Solve to show' does not mean just substituting in the value of  $n$ . Separation of variables is required. Also you should use  $\int \frac{1500}{n^2} dt = \int \dots dt$  as it is harder when you use  $\int \frac{1}{n^2} dt$  and have to carry the  $\frac{1}{1500}$  through the working.

**HINT**

Domain restriction includes practical restriction (time positive or zero) and functional restrictions.

**SLIP**

Be careful when separating the variables and with getting the formula transposition correct.

# 18 – Kinematics

*Kinematics is an application of Calculus to the motion of particles. There are some key words and definitions ideas that will help you work out what a question is asking for and what information is given. Throughout this sheet,  $x$  is used for position with units  $\text{cm}$ ,  $\text{m}$ ,  $\text{km}$ , etc.;  $v$  is used for velocity with units  $\text{cm/s}$ ,  $\text{m/s}$ ,  $\text{km/h}$  ( $\text{cms}^{-1}$ ,  $\text{ms}^{-1}$ ,  $\text{kmh}^{-1}$ ), etc.; and  $a$  is used for acceleration with units  $\text{cm/s/s}$ ,  $\text{m/s/s}$ ,  $\text{km/h/h}$  ( $\text{cms}^{-2}$ ,  $\text{ms}^{-2}$ ,  $\text{kmh}^{-2}$ ), etc.*



## TRICK

Remember:

- ☞ Initially means  $t = 0$
- ☞ At rest means  $v = 0$
- ☞ Uniform acceleration means acceleration constant
- ☞ Where? Means find  $x$ .
- ☞ When? Means find  $t$ .
- ☞ Distance travelled is not the same as the modulus of the displacement
- ☞ How far? Means find the distance.
- ☞ Average speed is  $\frac{\text{distance travelled}}{\text{time taken}}$
- ☞ Instantaneous means at that instant or time
- ☞ Rectilinear motion: motion in a straight line
- ☞ Retardation: slowing down

## THINGS TO KNOW

Constant (uniform) acceleration:

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \frac{1}{2}(u + v)t$$

To decide which formula to use, look at what is given and what is asked:  $v$ ,  $u$  and  $t$ ?  $s$ ,  $u$  and  $t$ ?  $v$ ,  $u$  and  $s$ ?

- ① A particle with initial velocity  $7 \text{ ms}^{-1}$  is subjected to an acceleration of  $-12 \text{ ms}^{-2}$ . What is its velocity after 2 seconds? What is its speed after 2 seconds?

- ② A rocket rises vertically at a constant speed of  $3 \text{ km}$  per sec. A protective tile breaks off when the rocket is  $h$  kilometres above the ground. Fifteen minutes later the tile hits the ground. Assuming that air resistance is negligible, find  $h$ .

## TRAP

Look at the direction of the initial velocity and the acceleration (which aren't zero) and make sure you have the correct units.

- ③ Consider this situation:

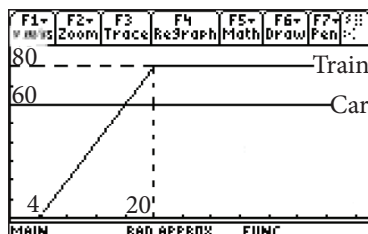
- Particle A is moving in a straight line with velocity of  $5 \text{ ms}^{-1}$  and acceleration  $2 \text{ ms}^{-2}$ .
- Particle B is moving in a straight line with velocity of  $-5 \text{ ms}^{-1}$  and acceleration  $2 \text{ ms}^{-2}$ .
- Particle C is moving in a straight line with velocity of  $5 \text{ ms}^{-1}$  and acceleration  $-2 \text{ ms}^{-2}$ .
- Particle D is moving in a straight line with velocity of  $-5 \text{ ms}^{-1}$  and acceleration  $-2 \text{ ms}^{-2}$ .

Which of the particles A, B, C, D in the situation described above are:

- Ⓐ speeding up?
- Ⓑ slowing down?
- Ⓒ moving forwards (in the positive  $X$ -direction)?
- Ⓓ moving backwards (in the negative  $X$ -direction)?



- 4 The graph shows the motions of a train and a car with velocity  $v \text{ ms}^{-1}$  plotted against time  $t \text{ s}$ . When does the train pass the car?

**HINT**

The vehicles must pass after the train reaches  $80 \text{ ms}^{-1}$ .

- 5 A lift runs between the ground floor and the top floor 55 metres up in a building. The lift can speed up at a rate of  $0.7 \text{ ms}^{-2}$  and slow down at a rate of  $1.5 \text{ ms}^{-2}$  or travel at a constant speed. The maximum speed allowed by law is  $3.7 \text{ ms}^{-1}$ .
- A Draw sketch graphs representing the position, velocity and acceleration of the lift at time  $t$ .
- B Find the shortest time that the lift could take to travel to the top floor without breaking the law. Give your answer correct to 4 decimal places.
- C What is the minimum time it would take if it broke the law?

**THINGS TO KNOW**

Displacement, velocity and acceleration are all vector quantities because they have direction as well as magnitude.

Displacement,  $x$ : change in position (vector)

Displacement from  $O$ : position (vector)

Instantaneous velocity:  $v = \frac{dx}{dt}$  (vector)

Instantaneous acceleration:  $a = \frac{dv}{dt}$ ,  $a = \frac{d^2x}{dt^2}$ ,  $a = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$  and  $a = v \frac{dv}{dx}$  (vector)

To decide which one to use, look at what is given and what is asked:  $v$ ,  $a$  and  $t$ ?  $x$ ,  $a$  and  $t$ ?  $v$ ,  $a$  and  $x$ ?

- 6 A particle moves in a straight line with velocity  $v \text{ ms}^{-1}$  at time  $t$  seconds given by  $v = \frac{4}{\sqrt{1-x^2}}$ ;  $0 < x < 1$ . Find the acceleration of the particle.

**SLIP**

$$a \neq \frac{dv}{dx}$$

- 7 A particle moves in a straight line with velocity  $v \text{ ms}^{-1}$  at time  $t$  seconds has acceleration  $a = \frac{dv}{dt} = \frac{2}{4-v}$ . Find the exact time taken for the velocity of the particle to change from  $0 \text{ ms}^{-1}$  to  $1 \text{ ms}^{-1}$ .

**TRAP**

Exact time is required.

- 8 The displacement of a particle,  $x \text{ m}$  from a fixed point  $O$  at time  $t \text{ s}$  is given by  $x = 1.5t + 4 \cos\left(\frac{t}{2}\right)$ ;  $t \geq 0$ . If the particle is moving in a straight line, find:

- A an expression for the velocity of the particle  $v \text{ ms}^{-1}$  at time  $t \text{ s}$
- B an expression for the acceleration of the particle  $a \text{ ms}^{-2}$  at time  $t \text{ s}$
- C the minimum value of the velocity of the particle  $v \text{ ms}^{-1}$ .

**TRICK**

Think about what it means to the acceleration when the velocity is a minimum. You may not need calculus.

# 19 – Kinematics and its applications

## THINGS TO KNOW



- ☞ Speed is not a vector quantity because it has magnitude but no direction.
- ☞ Acceleration:  $a = \frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$  (vector)
- ☞ Average velocity:  $\frac{\text{change in position}}{\text{change in time}} = \frac{\Delta x}{\Delta t}$  (vector)
- ☞ Average speed:  $\frac{\text{distance travelled}}{\text{time taken}}$  (scalar)
- ☞ Average acceleration:  $\frac{\Delta v}{\Delta t}$  (vector)

## TRAP

$x = f(t) \neq$  the equation of the path  
Instantaneous speed  $\neq$  Instantaneous velocity

- ① Find the velocity of a particle  $v \text{ ms}^{-1}$  in terms of the displacement  $x \text{ m}$  from a fixed point  $O$  if its acceleration is given by  $a = -6x$ . The particle starts from  $x = 2 \text{ m}$  with a velocity of  $0 \text{ ms}^{-1}$ .

## TRAP

There are two answers for the velocity and that is feasible. Think about why.

- ② A particle  $P$  moves in such a way that its position  $x$  at time  $t$  seconds is given by  $x(t) = 12t^3 - 18t^2 + 7t - 1$  for  $t \geq 0$ .

## HINT

Look at a graph.

- (A) Find an expression for the velocity  $v$  at time  $t$  seconds.  
(B) Hence find the maximum velocity in the first second and when it occurred.  
Give your answers correct to 2 decimal places.

- (C) If the average value of a function  $f(t)$  on the interval  $a \leq t \leq b$  is defined as  $\frac{1}{b-a} \int_a^b f(t) dt$ , find the average velocity of the particle in the first second.

## TRAP

You should show all your working clearly including factorisations, use correct notation, know your standard integrals or look them up, watch out for the constant of integration and working in degrees!  
That's plenty to remember including  $a = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$ .

- ③ A particle is moving such that its acceleration is given by  $\ddot{x} = 2x^3 + 8x$ . Its starting position is 3 metres to the right of the origin so  $x = 3$  when  $t = 0$ . Its initial velocity is  $13 \text{ ms}^{-1}$ .

- (A) Show that  $\dot{x} = x^2 + 4$ .  
(B) Hence find an expression for  $x$  in terms of  $t$ .

## HINT

Use the correct form of the acceleration by noticing that you have  $v$  and  $x$  and you want  $a$ . Also you will need a double angle trigonometric formula to simplify your answer.

- ④ A particle moves in a straight line in such a way that when it has a displacement  $x$  from the origin  $O$ , its velocity  $v$  is given by  $v = \cos 2x$ . Find the acceleration of the particle in simplest form.

- ⑤ A particle is moving such that its acceleration is given by  $\ddot{x} = 8x^3 + 12x^2 + 4x$ . Its position initially ( $t = 0$ ) is  $x = -2$  and its velocity is  $v = -4$ .

- (A) Show that  $v^2 = 4x^2(1 + x)^2$ .

## TRAP

Remember that  $a = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$  and  $a = v \frac{dv}{dx}$  as well as  $a = \frac{dv}{dt}$  and  $v = \frac{dx}{dt}$  so you need to choose the correct form for antidifferentiation. Don't ignore the  $+ c$ .



Ⓑ Hence or otherwise show that  $\int \frac{1}{x(1+x)} dx = -2t$ .

## HINT

Notice that the velocity is negative and you need to give the correct explanation for this. Also make sure that you demonstrate, using correct notation and complete steps, that you understand the question.

## SLIP

Watch your Algebra skills so you don't make any careless mistakes.

Ⓒ If  $\int \frac{1}{x(1+x)} dx = -\log_e \frac{1+x}{x} + c$ , find  $x(t)$  using the initial conditions.

Ⓔ A particle moves such that its position  $x$  at time  $t$  is given by  $x = t^3 - 9t^2 + 15t + 24$ ;  $t \geq 0$ . The graph of  $x$  against  $t$  is shown and  $x$  is in metres and  $t$  is in seconds.

Ⓐ Find expressions for:

Ⓘ velocity  $v$

Ⓙ acceleration  $a$ .

Ⓑ Sketch graphs of  $v$  against  $t$  and  $a$  against  $t$  on the axes provided.

Ⓒ Find the position, velocity and acceleration initially.

Ⓓ When and where is the particle at rest?

Ⓔ When and where is the velocity constant?

Ⓕ When is the particle

Ⓙ speeding up while moving in the positive direction?

Ⓙ speeding up while moving in the negative direction?

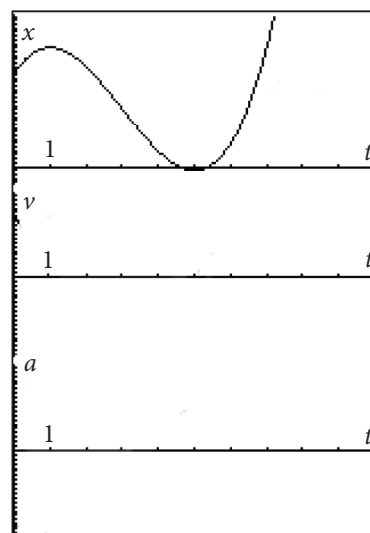
Ⓙ slowing down while moving in the positive direction?

Ⓙ slowing down while moving in the negative direction?

Ⓕ The graph shows the velocity  $v$  cm per second at time  $t$  seconds of a particle as it moves along the  $x$ -axis according to the rule  $v = t^3 - 6t^2 + 5t$ . The particle is initially 3 centimetres to the right of the origin,  $O$ .

Ⓐ Find an expression for  $x$ , the position of the particle, as a function of time.

Ⓑ Find the exact distance travelled by the particle in the first 2 seconds.

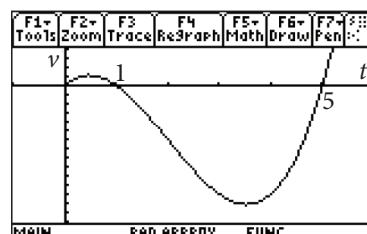


## HINT

Think of these where  $v$ 's direction is the direction of motion and  $a$ 's direction is like the direction of a force pushing.

## TRAP

Distance travelled is different from change in displacement if the particle changes direction.



# 20 – Kinematics in real situations

Practical applications of kinematics frequently come from the 'real world' and you need practice in interpreting the words into mathematical symbols.



- ① When a rocket is fired vertically upwards from the surface of the Earth, its velocity  $v$  at a distance  $x$  from the centre of the Earth is given by solving the differential equation:  $v \frac{dv}{dx} = -\frac{k}{x^2}$  where  $k$  is a positive constant.

- Ⓐ Show that if the initial velocity of the rocket at the surface of the Earth is  $v_0$  and  $R$  is the radius of the Earth, the solution of this differential equation is  $v^2 = v_0^2 + 2k \left( \frac{1}{x} - \frac{1}{R} \right)$ .

- Ⓑ Find  $v_0$ , the initial velocity of the rocket (to 1 decimal place), if the rocket gets to a maximum height of  $R$  from the surface of the Earth. Use  $R = 6.37 \times 10^3$  units and  $k = 3.98 \times 10^5$  units.

## HINT

Don't forget the  $+C$  and make sure you understand the meaning of the particular solution. Is the differential equation in the easiest form for solution? Why is  $x = 2R$  when the rocket is a distance  $R$  above the surface of the Earth? This means you want to get  $v_0$  when  $x = 2R$ .

- ② A sky diver falling under gravity in air is subjected to two forces, gravity (acceleration  $g = 9.8 \text{ ms}^{-2}$ ) and drag,  $D$ .

- Ⓐ If the drag is proportional to the square of the velocity  $v$  of the diver with constant of proportionality  $k$ , write down an expression for the drag.

- Ⓑ Hence complete the equation of motion of the diver where  $x$  metres is the displacement in metres at time  $t$  seconds after jumping from point  $O$ .

$$\ddot{x} = \underline{\hspace{2cm}}$$

- Ⓒ The terminal velocity of the diver is reached when he is no longer accelerating. Show that the terminal velocity is given by  $v = \sqrt{\frac{g}{k}}$ .

- Ⓓ If the terminal velocity for this diver is  $54 \text{ ms}^{-1}$ , find the exact value of  $k$ .

- Ⓔ Show that  $\frac{dx}{dv} = \frac{v}{g - kv^2}$ .

- Ⓕ Hence show by integration that  $v^2 = \frac{g}{k} - e^{-2kx} + C$  where  $C$  is the constant of integration.

- ③ A particle  $P$  falls at time  $t = 0$  from the inside of the rim of a wheel of radius  $r$  rotating at an angular velocity  $\omega$ . For  $\theta$  the angle that  $OP$  makes to the vertical, the position of  $P$  at time  $t$  seconds after it begins to fall is given by:

$$x(t) = r \sin \theta - r \omega t \cos \theta$$

$$y(t) = r \cos \theta + r \omega t \sin \theta - 4.9t^2$$

- Ⓐ By considering the function  $x(t)$ , find the time  $t$  when the particle again makes contact with the rim of the wheel at point  $Q(0, -r)$ .

- Ⓑ It can be shown that  $\cos \theta = \frac{r\omega^2}{9.8}$ . (Do not prove or derive this.)

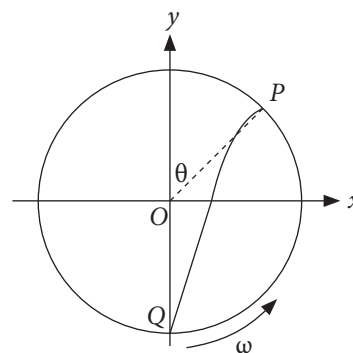
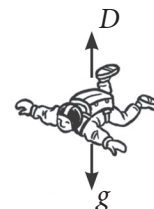
By considering the function  $y(t)$  and the result from part a, show that  $2\cos^3 \theta + 3\cos^2 \theta - 1 = 0$ .

- Ⓒ Show that a solution to the equation  $2\cos^3 \theta + 3\cos^2 \theta - 1 = 0$  for  $0 \leq \theta \leq \frac{\pi}{2}$  is  $\theta = \frac{\pi}{3}$ .

- Ⓓ Hence find the angular speed of the wheel,  $\omega$ , for a wheel of radius  $0.60 \text{ m}$ . Give your answer correct to 2 decimal places.

## HINT

What value of  $x$  should you use in the first part of the question? You will need a clear head to systematically eliminate  $t$ ,  $\theta$  and  $r$  to get the equation in  $\cos \theta$ . Be careful of your algebraic manipulations. The final parts of the question just need substitutions.



# 21 – Perms, combs and probability

Check with your teacher to make sure that these topics are on your syllabus.  
Perhaps you studied them last year as well.



- ① Alexandra (A) and Brigitte (B) are taking turns tossing a coin to see who is first to get 5 heads. The results are listed by who scores each head: for example, ABBAABAA means Alexandra was the first to get 5 heads.

- Ⓐ Why are there 5 different ways for Alexandra to win if Brigitte gets only one head?
- Ⓑ In how many ways could the results of a completed game be recorded?

## TRICK

Try listing the possible outcomes and write a sentence of explanation to say 'why'. Remember that A or B won when there were 5, 6, 7, 8 or 9 heads recorded.

- ② A four-person tennis team is to be selected from 8 boys and 4 girls.

- Ⓐ In how many ways can the team be chosen?
- Ⓑ What is the probability that the team will be all girls?

## HINT

Does it matter about being boys on girls in the first part? And you probably need to use your answer to the first part in working out the second part.

- ③ Ⓐ In how many ways can  $n$  counters be placed in two different cups so that neither cup is empty?

- Ⓑ In how many ways can 4 counters be placed in three different cups so that no cup is empty?

## TRICK

Be careful if you are adding binomial coefficients that you don't include  $\binom{n}{0}$  and  $\binom{n}{n}$ . Of course the second part of the question is connected with the first part so don't just count; you may miss some of the combinations.

- ④ Lilli selects five cubes at random from a bag containing 3 red, 3 orange, 3 yellow, 3 green and 3 blue cubes.

- Ⓐ How many selections contain one of each colour?
- Ⓑ How many different 'castles' can she make by stacking the cubes 3 at a time on top of each other?

## HINT

One question is permutations and one is combinations. You need to ask the question: 'Is the order important?'

- Ⓒ How many 'castles' can she make by stacking 2, 3, 4 or 5 cubes on top of each other?

- ⑤ If  $\binom{n-1}{k-1} + \binom{n-2}{k-1} + \binom{n-3}{k-1} + \dots + \binom{k-1}{k-1} = \binom{n}{k}$ ,  
show that  $n\binom{n-1}{k} = (k+1)\binom{n}{k+1}$ .

## TRICK

Do not work with both sides of the identity. Say what you are doing at each step.

## THINGS TO KNOW

Some of the following may be on your formula sheet but you should still know the rules so you recognise them when they come up.

$$\Pr(A) = 1 - \Pr(A')$$

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

If  $X$  has a binomial distribution,  $\Pr(X=x) = {}^nC_x p^x (1-p)^{n-x} = \binom{n}{x} p^x (1-p)^{n-x}$ , mean or expected value is  $np$  and variance is  $np(1-p)$ .



- 6 In an orienteering event, runners compete in teams of two or four runners. The probability that an individual runner will finish the course in less than 15 minutes is  $p$  and in more than 15 minutes is  $q = 1 - p$ . A team scores full points if at least half the runners finish the course in less than 15 minutes.

- A Find an expression in terms of  $p$  **only** for the probability that a two-person team

I has no runners who take less than 15 minutes to finish the course

II will score full points.

- B Find an expression for the probability in terms of  $p$  **only** that a four-person team

I has at least three runners who take more than 15 minutes to finish the course

II will score full points.

- C Hence or otherwise find the range of values of  $p$  for which the two-person team is more likely to score full points rather than the four-person team. Give your answer correct to 3 decimal places.

- 7 Given that  $\Pr(X = 0) + \Pr(X = 1) + \Pr(X = 2) + \Pr(X = 3) + \Pr(X = 4) + \Pr(X = 5) = 1$ , write down each of the following in the form  $1 - (\text{a sum of probabilities})$ .

- A the probability of at least 3  
B the probability of at most 3  
C the probability of no more than 3  
D the probability of no less than 3

- 8 The committee of management of my apartment building comprises 10 owners of which I am one. Committee meetings are held 5 times per year and at each meeting a chairperson is chosen at random.

- A What is the probability, correct to 3 decimal places, that in the first year I will be chosen to chair the meeting at least once?  
B Show that in the first 4 years, I have a better chance of chairing exactly twice than of chairing exactly once.  
C For how many years must I stay on the committee so that I have a better chance of chairing exactly three times than of chairing exactly twice?

- 9 Mike takes a 6-question multiple choice test every week of term. Each question has four choices and he guesses the answer to each question. He earns 10 marks if he picks more than half the correct answers and no marks otherwise.

- A What is the probability that Mike picks the correct answer to any given question?  
B Show that the probability that Mike earns 0 marks in any given test is 0.9624, correct to 4 decimal places.  
C Hence find the probability that Mike earns a total of no marks during a 10 week term; that is, he earns no marks every week of the term. Give your answer correct to 4 decimal places.  
D Find the probability that Mike earns at most 20 marks during the 10 week term. Give your answer correct to 4 decimal places.

### THINGS TO KNOW

- Read the questions carefully and always check the form of the answer specified.
- You need to be careful of algebraic errors in the substitutions between  $p$  and  $q$  and the simplifications of the answers.
- Remember that 'hence' is a clue to the method to use but it also means that if your answers to the previous parts are complicated your working in this part will be made more difficult.
- Watch out for careless answers like two roots to equations or negative probabilities.

### SLIP

Careless errors are easy to make in this question so watch the accuracy in your manipulations.

### SLIP

Using Binomial distribution formulae when numbers are in decimal form can lead to plenty of errors so make sure all your brackets are in place and that you choose the correct values of  $n$ ,  $p$  and  $q$ .

# 22 – Checklist for success

Everyone works differently so you should change this checklist to suit yourself and add some other things if you like. The important thing is to have a plan. That way you are less likely to put off the things you find hard. Finally, don't stress about a problem, do something about it: spend more time on it or see the teacher.



| Revision time                                                                                                                                            | Done<br>✓ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| I have tried all my homework and asked the teacher about any problem areas.                                                                              |           |
| I have made a revision plan by dividing the course into blocks of time and have worked hard to stick to it.                                              |           |
| My revision plan includes recreation and relaxation.                                                                                                     |           |
| My revision plan includes a couple of free times in case I need to catch up because of sickness, etc.                                                    |           |
| I have altered my revision plan if I need to (sickness, etc.) immediately I needed to and not waited until the end.                                      |           |
| I have helped myself learn by making my own notes of important points with examples, and organising a summary of the work in my own words.               |           |
| I have thought about key ideas and methods, learned formulae, made summary tables and/or mind maps.                                                      |           |
| I have ticked off my list of things as I have revised and always look at how much I've done rather than how much I have left to do.                      |           |
| I have rewarded myself each time I reach a target by going to the refrigerator or watching TV or calling a friend.                                       |           |
| I have moved to the next topic knowing that I can talk to myself about the definitions of terms, the key points, real examples, formulae and graphs.     |           |
| I know what sorts of questions I am likely to be asked: short answer/multichoice/...                                                                     |           |
| I have practised each type of exam question (theory, skills, interpret, analyse)                                                                         |           |
| I know what to do with questions with several parts.                                                                                                     |           |
| I have worked through at least the last 2 year's exams against the clock in the required time using only the tools I am allowed to use in the real exam. |           |

|                                                                                                                                                                                                            |                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| I have had practice working for 2–3 hours without a break like I will need to do in the exam.                                                                                                              |                   |
| I have worked out how long each type of question should take by dividing the total exam time for each type by the number of questions.                                                                     |                   |
| I have checked that I have an approved calculator and spare batteries.                                                                                                                                     |                   |
| I know what will be on the formula sheet provided.                                                                                                                                                         |                   |
| I have checked that I have pencils, pens, highlighters, ruler, eraser and other mathematical equipment.                                                                                                    |                   |
| I know what written notes (if allowed) and other things I can take into the exam.                                                                                                                          |                   |
| I have a watch or clock I can use.                                                                                                                                                                         |                   |
| I have checked when the exam is on and where.                                                                                                                                                              |                   |
| I have worked out how long it takes to travel to the exam and how I will travel.                                                                                                                           |                   |
|                                                                                                                                                                                                            |                   |
|                                                                                                                                                                                                            |                   |
| <b>The night before the exam</b>                                                                                                                                                                           | <b>Done<br/>✓</b> |
| I have put together what I need for the exams: calculator, spare batteries, pencils, pens, highlighters, written notes (if allowed), ruler, eraser and other mathematical equipment, and a watch or clock. |                   |
| I have put out the clothes I need.                                                                                                                                                                         |                   |
| I have not done any more questions or tried to 'cram', but have just read through my notes to refresh my memory and looked at questions I have already done correctly.                                     |                   |
| I have checked the length of the paper and worked out how long I should spend on each question.                                                                                                            |                   |
| I keep telling myself how much I know and understand.                                                                                                                                                      |                   |
| I have double-checked when the exam is and where.                                                                                                                                                          |                   |



|                                                                                                                                                                                                                                              |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| I have double-checked my alarm clock (mechanical and human) to give myself plenty of time.                                                                                                                                                   |                    |
| I have had a shower or a bath to relax and gone to bed at a reasonable time.                                                                                                                                                                 |                    |
|                                                                                                                                                                                                                                              |                    |
|                                                                                                                                                                                                                                              |                    |
| <b>Exam morning</b>                                                                                                                                                                                                                          | <i>I will</i><br>✓ |
| Have a good breakfast or lunch before I leave home.                                                                                                                                                                                          |                    |
| Collect all my exam equipment.                                                                                                                                                                                                               |                    |
| Leave home in plenty of time to get there.                                                                                                                                                                                                   |                    |
| Check that I am at the right room and the starting time for the exam.                                                                                                                                                                        |                    |
|                                                                                                                                                                                                                                              |                    |
|                                                                                                                                                                                                                                              |                    |
| <b>During the exam – good habits</b>                                                                                                                                                                                                         | <i>I will</i><br>✓ |
| Use my reading time to read the paper carefully, set times for different questions/ sections and to work out which questions I will do first and which ones I think are harder and need to be done later in the exam.                        |                    |
| Spend the correct amount of time on each question and go on to the next one even if I haven't quite finished because I know I will have most of the marks for the answer by then anyway.                                                     |                    |
| Return to unfinished questions after I have attempted all the questions I need to.                                                                                                                                                           |                    |
| Try to write something down if I have a mental block because I know that writing down a few relevant facts, equations or formulae, will probably jog my memory enough to get started (and I won't get any marks for a blank sheet of paper). |                    |
| Never leave out answers to multiple choice or short answer questions. I will have a go since there are no marks lost for wrong answers and I won't get marks for gaps!                                                                       |                    |
| Always try to ask myself 'What does the examiner want?'                                                                                                                                                                                      |                    |
| Answer the question asked, not the one I think was asked.                                                                                                                                                                                    |                    |

|                                                                                                                                                                  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Use the method the examiner has said to use (if specified).                                                                                                      |  |
| Give the answer in the form the examiner has asked for (if specified).                                                                                           |  |
| Reread the question when I have finished to make sure I have answered the question                                                                               |  |
| Be extra careful with questions that I find easy because I may not think carefully enough.                                                                       |  |
| Watch out for key words like 'hence' (use something done or given in an earlier part) and 'show' (prove or demonstrate).                                         |  |
| Not use the calculator when the question asks for exact answers.                                                                                                 |  |
| Use estimation to check my answer is sensible.                                                                                                                   |  |
| Use my calculator wisely, making sure that I put in brackets where I need them, that I am in the right mode (radian/degree).                                     |  |
| If I have a graphing calculator, always adjust the window so I can see the key features of any graph.                                                            |  |
| If I have a graphing calculator, use the CALC and not the TRACE button.                                                                                          |  |
| Take care to not make arithmetic and algebraic mistakes.                                                                                                         |  |
| Round my numerical answer at the end rather than during the working.                                                                                             |  |
| Show all my working.                                                                                                                                             |  |
| Make sure my answers are written legibly and logically.                                                                                                          |  |
| Make sure I have put units on the answers if necessary.                                                                                                          |  |
| Above all, take the chance I have been given to show the examiner what I know and understand and will feel confident that I have done my best, whatever my mark. |  |

# Answers

## 1 – Some stuff is assumed knowledge

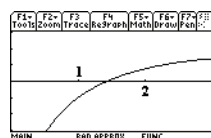
- 1** **A**  $x \leq 2$  or  $(-\infty, 2]$  **B**  $R$  **C**  $-2 < x \leq 2$  or  $(-2, 2]$   
**2**  $y = \pm\sqrt{16 - x^2}$   
**3**  $-2$   
**4**  $y = \frac{x^2}{x-1}$   
**5**  $4$   
**6**  $(2 \times 4 - 1) + (2 \times 5 - 1) + (2 \times 6 - 1) + (2 \times 7 - 1) = 40$   
**7** **A**  $n$  **B**  $\frac{n(n+1)}{2}$ ; series approaches infinity  
**8** **A**  $(\frac{2}{3})^{n-1}$  **B**  $3[1 - (\frac{2}{3})^n]$ ; series approaches 3  
**9**  $8b^3 - 36b^2 + 54b - 27$   
**10**  $81x^2 - 162x + 81$   
**11**  $-96xy^6$   
**12**  $5103x$   
**13**  $2a^2 - \frac{1}{3}a + \frac{8}{9}$  with remainder  $-\frac{25}{9}$

## 2 – Best ways to factorise

- 1** **A**  $3(x+2y-9z)$  **B**  $4(5p-q)(5p+q)$  **C**  $(a-\sqrt{15})(a+\sqrt{15})$   
**D**  $8(x+2y)(x^2-2xy+4y^2)$  **E** No real factors  
**2** **A**  $(2x-3)^2$  **B**  $(2y-3)(y+3)$  **C**  $(x-4-2\sqrt{3})(x-4+2\sqrt{3})$   
**D**  $(a-\sqrt{5})(a+\sqrt{5})(a-\sqrt{7})(a+\sqrt{7})$  **E**  $(b+3)(b^2-3b+5)$   
**3** Yes; discriminant is positive; perfect square so rational (fractional) factors  
**4** **A**  $(b-4)(b^2+4b+16)$  **B**  $3(x-\frac{5}{6}-\frac{\sqrt{61}}{6})(x-\frac{5}{6}+\frac{\sqrt{61}}{6})$   
**C**  $(x+2y+3z)(x+2y-3z)$  **D**  $(y+1)(y-3)(y+4)$   
**5**  $a=3, b=14, c=8; (y-2)(3y^2+14y+8); (y-2)(3y+2)(y+4)$   
**6**  $r(\sqrt{p}r-\sqrt{q})(\sqrt{p}r+\sqrt{q})$   
**7** **A** Quotient is  $x$ ; remainder is  $x-1$  **B**  $k=(-1)^{n+1}$  **C**  $n$  even

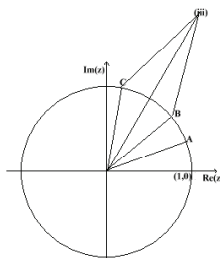
## 3 – Good algebra skills are a 'must'

- 1**  $\pm\sqrt{16-h^2}$   
**2**  $y = \frac{7e^2+1}{64}$   
**3**  $6$   
**4** No answer given  
**5**  $y = -4.9(\frac{x}{100})^2 + x + 4000$   
**6**  $a=2, b=-6$   
**7** No answer given  
**8**  $A=2; B=-2$   
**9** See graph at right;  $f(1) = 4\log_e x - x = -1$  and  $f(2) = 4\log_e 2 - 2 \approx 0.77$   
**10**  $9x\sqrt{4-x^2} + 6\sqrt{x} + \frac{1}{\sqrt{4-x^2}}$   
**11** No answer given  
**12** No answer given

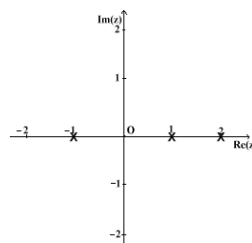


## 4 – De Moivre's Theorem

- 1**  $\text{cis}(-\pi), \text{cis}(-\frac{\pi}{3}), \text{cis}(\frac{\pi}{3})$ ; note that  $\text{cis}(\pi) = \text{cis}(-\pi)$ , etc.  
**2**  $3$   
**3**  $1+i, -1+i, -1-i, 1-i$ ; yes  
**4**  $f(\text{cis}\frac{3\pi}{2}), f(\text{cis}3\pi), f(\text{cis}\frac{9\pi}{2})$  and  $f(\text{cis}6\pi)$   
**5** **A**  $B$  **B**  $C$  **C** See diagram at right  
**D** **E** No answer given



- F** (i)  $-\frac{\pi}{3}, 0, \frac{\pi}{3}$  (ii)  $-1, 2, 1$  (iii) See diagram below



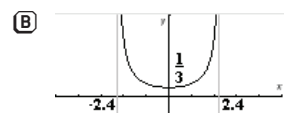
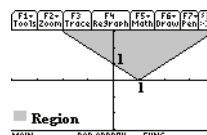
- 6** **A** See hint **B** Simplest answer is 1  
**7**  $-\frac{1}{2}$   
**8** No answer given  
**9** **A**  $\text{cis}\frac{3\pi}{4}$  **B**  $\text{cis}\frac{n\pi}{8}$   
**10** **A** (i)  $\frac{1-\sqrt{3}}{2} + i\frac{1+\sqrt{3}}{2}$  (ii)  $2\text{cis}(\frac{\pi}{3})$  **B**  $\sqrt{2}\text{cis}(\frac{\pi}{12})$   
**C**  $\frac{1-\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{2}-\sqrt{6}}{4}$

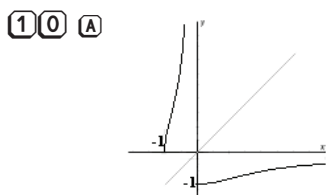
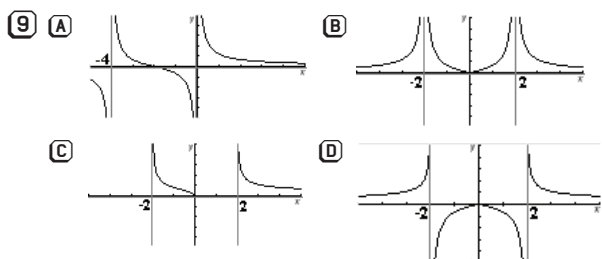
## 5 – Complex numbers and algebra

- 1**  $\frac{-i \pm \sqrt{3}}{2}$   
**2**  $3$  **A**  $z=2i, 2i, -1$  (repeated solution) **B**  $-1, \frac{1}{2} + \frac{\sqrt{3}}{2}i, \frac{1}{2} - \frac{\sqrt{3}}{2}i$   
**3** **A**  $4$  **B**  $-2, i$   
**4**  $-3i$   
**5** Betty  
**6** No answer given  
**7** **A**  $P(-2)=0$  **B**  $x^3-7x^2+16x-10$  **C**  $x^2-6x+10$   
**D**  $3+i, 3-i$  **E**  $1, -2, 3+i, 3-i$   
**8** **A**  $(x-1)(x+1)(x^2+4)$  **B**  $(x-1)(x+1)(x-2i)(x+2i)$   
**9**  $(x + \frac{3}{2} - \frac{\sqrt{5}}{2})(x + \frac{3}{2} - \frac{\sqrt{5}}{2})$   
**10**  $(x + \frac{3}{2} - \frac{\sqrt{3}}{2}i)(x + \frac{3}{2} - \frac{\sqrt{3}}{2}i)$   
**11**  $a=1, b=0, c=13, d=-34$   
**12** None  
**13**  $-7+24i$

## 6 – Functions and relations

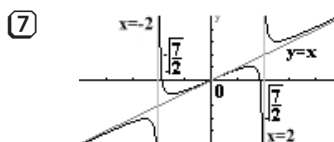
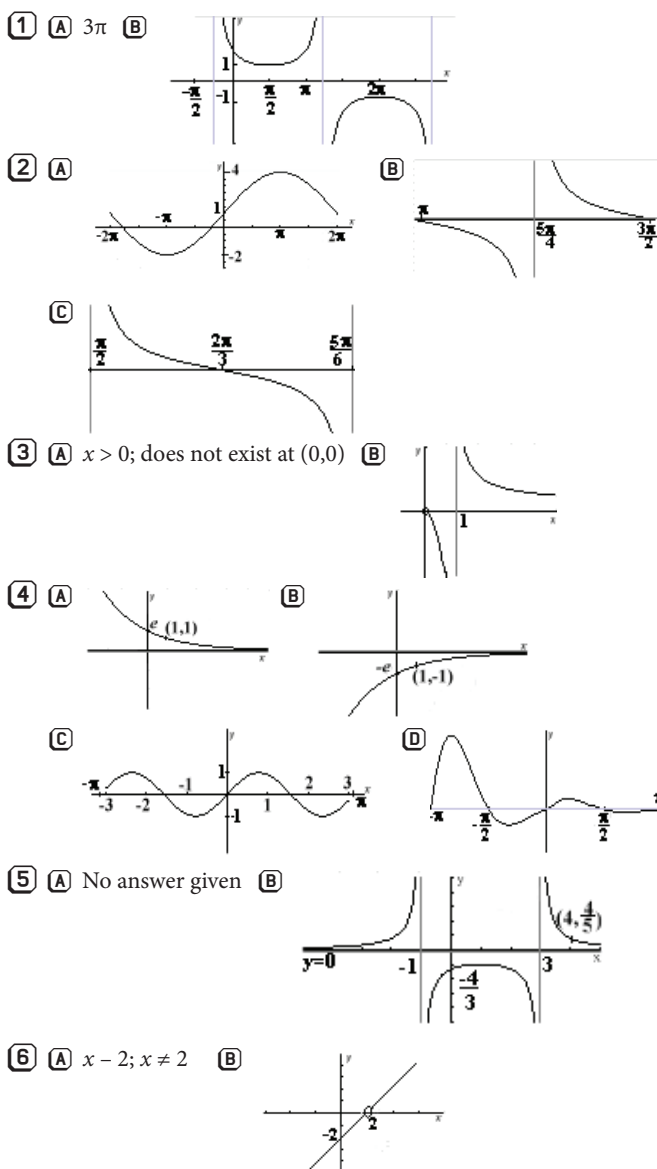
- 1** **A**  $\sin(-x^3) = -\sin(x^3)$  **B**  $0$   
**2** (i) T; even powers of polynomial gives odd powers after differentiation (ii) F; even powers of derivative gives odd power of polynomial but also get +C  
**3**  $f(0)=1, f(1)=2, f(2)=5, f(3)=10$ ; increasing  
**4**  $f^{-1}(x) = 4\sin x$   
**5** **A**  $(-2, 2)$  or  $-2 < x < 2$  **B**  $[0, 2)$  or  $0 \leq x < 2$   
**6** No answer given  
**7**





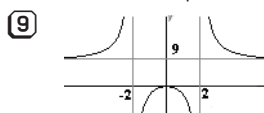
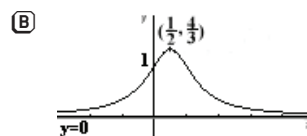
(B)  $-1 \leq x < 0$  or  $[-1, 0)$  (C)  $f^{-1}(x) = \sqrt{\frac{-4}{x} - 4} = 2\sqrt{\frac{-1-x}{x}}$   
 where  $-1 \leq x < 0$  (D) (i)  $f^{-1}(-x) = \sqrt{\frac{4}{x} - 4}$  where  $0 < x \leq 1$ ;  
 $-f(x) = \frac{4}{x^2 + 4}$  where  $x \geq 0$  (ii)  $\frac{4}{x^2 + 4} = \sqrt{\frac{4}{x} - 4}$   
 (iii) No answer given

## 7 – Graphs of functions and relations



8 (A)

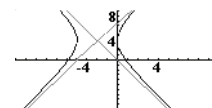
|        |        |     |    |     |    |      |      |                       |                       |                       |
|--------|--------|-----|----|-----|----|------|------|-----------------------|-----------------------|-----------------------|
| Denom. | 0      | <+1 | +1 | >+1 | -1 | max. | min. | $\rightarrow 0^+$     | $\rightarrow 0^-$     | $\rightarrow +\infty$ |
| $f(x)$ | undef. | >1  | +1 | <1  | -1 | min. | max. | $\rightarrow +\infty$ | $\rightarrow -\infty$ | $\rightarrow 0$       |



10 (B)  $B^2 < -4C$

## 8 – Coordinate geometry and conic sections

1 (-2,3)  
 2  $(\frac{52}{5}, \frac{11}{5})$  or (10.4,2.2)  
 3 No answer given  
 4  $1 + \sqrt{8}$  ( $x = 0$ )  
 5 (A)  $a = -2$ ;  $\frac{(x+2)^2}{4} - \frac{(y-4)^2}{16} = 1$  (B) scale shown; asymptotes  $y = -2x$  and  $y = -2x + 8$ ; centre (-2,4); vertices (-4,4), (0,4)  
 6 (A) No answer given (B)  $e \geq \sqrt{2}$   
 7 (A) Substitution (B) On the directrix  
 (C)  $(ae, b\sqrt{1-e^2})$  (D)  $AF = b\sqrt{1-e^2}$ ;  $FT = \frac{a}{e}(1-e^2)$ ;  
 $\tan^{-1}(\frac{eb}{a\sqrt{1-e^2}})$  (E)  $\frac{ab}{2e}(1-e^2)^{\frac{3}{2}}$



## 9 – Geometry in a circle

1 No answer given  
 2 No answer given  
 3 Possible reasons: (A) Opposite angles supplementary (B) AC a diameter because of angles in a semicircle (C) All radii  
 4 Possible reasons: alternate angles, bisection and exterior angle  
 5 4 units  
 6 (A) No answer given (B) No answer given  
 7 Possible reasons: (A) Angles in alternate segments (B) Congruent triangles  
 8 (A)  $90^\circ$ ; possible reason: opposite angles of cyclic quadrilateral and angles at the centre and angles standing on the same arc (B) Possible reason: BC must be a diameter because of part a.  
 9 Possible reasons: (A) Opposite angles supplementary (B) Angles in a circle standing on the same arc (C) Angle sums

## 10 – Vector algebra and geometry

1 Doesn't exist, so false  
 2 (A)  $p = 1$ ;  $q = 4$  (B) 43  
 3  $i + 4j$



- 4**  $-4 - 2i$   
**5** **A** No answer given **B** No answer given **C**  $c[22, 5, -1]$   
**6**  $3a + 4b = 20$   
**7**  $8i$ ;  $10j$  **B** Possible reason: dot product zero  
**8** **A** Possible reason: angle in a semicircle **B**  $(2, 0)$   
**9** **A** **B** No answer given  
**10** **A** (i)  $\vec{AB} = \vec{b} - \vec{a}$  (ii)  $\vec{OP} = \frac{1}{3}\vec{a}$   
**B** Possible reason: scalar multiple **C**  $\frac{\sqrt{3}}{4}ab$ ;  $\frac{\sqrt{3}}{9}ab$

## 11 – Differentiation

- 1** No answer given  
**2**  $6x - 3x^2$   
**3**  $n(1+x)^{n-1}$   
**4** **A**  $\frac{1}{\sqrt{2x}}$  **B**  $\frac{1}{\sqrt{2x}} \cdot \frac{1}{1+2x}$   
**5** **A**  $\frac{4x}{\sqrt{1-4x^4}}$  **B**  $\frac{-6}{\sqrt{1-4x^4}}$   
**6** **A** **B** (i)(ii) No answer given (iii)  $2e^x \cos x$   
**7** **A**  $-1 < x < 1$  **B**  $-a < x < a$ ;  $a > 0$   
**8** No answer given  
**9**  $a = -1$ ;  $b = -1$   
**10** **A**  $P(x) = (x+1)Q(x)$ ;  $P'(x) = (x+1)Q(x) + R(x)$   
**B** No answer given

## 12 – Gradients, tangents and normals

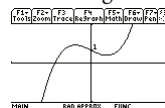
- 1** 23  
**2** 0  
**3** **A**  $\frac{2}{e}$  **B**  $y = \frac{2}{e}x$   
**4**  $y = -4x$   
**5**  $\pm 2$   
**6**  $\tan^{-1}\left(\frac{f(0.507) - f(0.505)}{0.002}\right) \approx 54^\circ$   
**7** **A** No answer given **B** No answer given **C** 0.990

## 13 – Sketch graphs and calculus

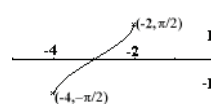
**1**

|                                                              | $f'(x) < 0$<br>Gradient<br>negative | $f'(x) = 0$<br>Gradient<br>zero | $f'(x) > 0$<br>Gradient<br>positive |
|--------------------------------------------------------------|-------------------------------------|---------------------------------|-------------------------------------|
| $f''(x) < 0$<br>Rate of<br>change of<br>gradient<br>negative |                                     |                                 |                                     |
| $f''(x) = 0$<br>Rate of<br>change of<br>gradient<br>zero     |                                     |                                 |                                     |
| $f''(x) > 0$<br>Rate of<br>change of<br>gradient<br>positive |                                     |                                 |                                     |

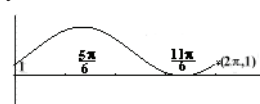
- 2** **A** 0 **B** 0; minimum, not an inflection point  
**3** **A** No answer given **B** Possible reason: the graph looks like this:



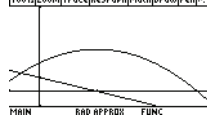
- 4** **A** No answer given **B** No answer given **C** Possible reason: 1 : 1  
**5** **A** 1 **B**



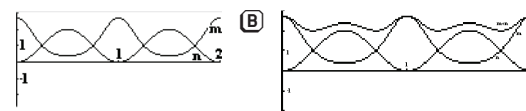
- 6** **A**  $f'(\alpha) = \sqrt{3} \cos \alpha + \sin \alpha$  **B**  $\frac{2\pi}{3}, \frac{5\pi}{3}$   
**C**



- 7**



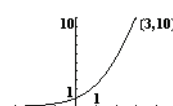
- 8** Local minimum  $x = \frac{\pi}{2}$ , local maximum  $x = \frac{3\pi}{2}$ , no inflection points  
**9** **A**



- C**  $m$ : 1;  $n$ : 1 **D** Maxima at:  $m$ : 0,1,2; value  $e$ .  $n$ : 0.5,0.5; value 2. Minima at:  $m$ : 0.5,0.5; value  $\frac{1}{e}$ ;  $n$ : 0,1,2; value 0  
**E** Values where maxima and minima occur are reversed.

## 14 – Maximum and minimum of functions

- 1** **A**  $e^{1-x}(2 \cos 2x - \sin 2x)$  **B** No answer given  
**C**  $2x = n\pi + \tan^{-1}2$  or  $x = \frac{n\pi}{2} + \frac{1}{2}\tan^{-1}2$  or  $x = \frac{n\pi}{2} + 0.5536$   
**2** **A**



- 3** **A**  $x = \frac{h+k}{2}$  **B** No answer given **C** No answer given  
**4** **A**  $f'(x) = \frac{3a^2}{2} \left( \frac{1 - \sqrt{3} \cos x}{\sin^2 x} \right)$  **B** 0.955 **C** Possible reason: solving  $f'(x) = 0$  and substituting back in  $f(x)$   
**5** No answer given  
**6**  $2\sqrt{15}$  units  
**7** **A** 55 **B** 5 **C** 1 min **D** 0.3155 min  
**E**  $0.6845 - 0.3155 = 0.369$  min

## 15 – Rates of change

- 1** 9.01; exact rate is 9  
**2** **A**  $20 \sin(\pi t)$  mms<sup>-1</sup> **B** 0 mms<sup>-1</sup> **C** 44.4 mms<sup>-1</sup>  
**3** **A**  $150x^{\frac{1}{2}}$  **B**  $\frac{dx}{dt} = \frac{dx}{dy} \cdot \frac{dy}{dt}$  **C**  $\frac{dx}{dt} = \frac{4}{150x^2}$  **D**  $\frac{1}{150}$   
**4**  $\frac{10}{\pi}$  cms<sup>-1</sup>  
**5** **A** No answer given **B** -\$1080 per week (falling)  
**6** **A** No answer given **B** No answer given **C** Possible reasons:  $x$  becomes smaller as the radius becomes smaller;  $\sin \theta$  is negative  
**7** **A**  $x = \sqrt{25 - h^2}$  **B**  $\frac{2}{3}$  ms<sup>-1</sup>



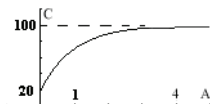
- 8 (A) Possible method: use chain rule (B) (i)  $\frac{8\pi}{81k}R^3$  (ii)  $\frac{28\pi}{81k}R^3$   
(iii)  $\frac{2\pi}{3k}R^3$

## 16 – Differential equations

- 1 No answer given  
2 No answer given  
3 (A)  $y = +\sqrt{2x+4}$  (B)  $y = \cos x$   
4 (A) (C) No answer given (D)  $x = 3 - \frac{1}{t+1}$  (E)  $\frac{d^2x}{dt^2} = -2(3-x)^3$   
5 (A)  $-\frac{1}{n} \log_e(m - nA)$  (B) No answer given (C) 6.46  
6 No answer given  
7  $\frac{1}{3} \log_e(1 - 3x)$ ;  $x < \frac{1}{3}$   
8  $u^2 = x^2 - 25$ ;  $y = \frac{1}{3}(x^2 - 25)^{\frac{3}{2}} - 76$   
9  $k = 4$   
10 (A) (B) No answer given

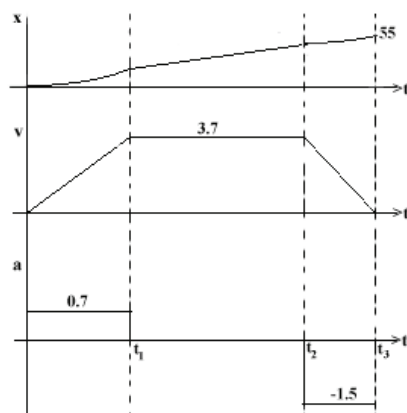
## 17 – Practical problems involving differential equations

- 1 (A) Surrounding temperature  $S$ ; the negative indicates cooling/temperature falling (B)  $40.5^\circ\text{C}$   
2  $5^\circ\text{C}$   
3  $\frac{dT}{dt} = 25 - \frac{1}{4}(T - 20)$ ;  $t = 0$ ,  $T = 20^\circ$   
4 (A) No answer given (B) 36 (C) 0.1386;  $T = 36e^{-0.1368t} - 1$   
(D)  $8^\circ\text{C}$   
5 (A) No answer given (B) 38.5 minutes  
6  $\frac{dx}{dt} = \frac{(3-x)}{25}$   
7 (A) No answer given (B) Because  $t$  must be positive,  $t = 15$  gives undefined function and  $t > 15$  would make  $n$  negative.  
8 (A) No answer given (B) asymptote  
 $C = 100$  (C)  $\frac{20}{C(20-C)}$   
(D) No answer given  
(E)  $k = \log_e 19$



## 18 – Kinematics

- 1  $-17 \text{ ms}^{-1}$ ;  $17 \text{ ms}^{-1}$   
2 1269 ( $u = -3000$ ;  $a = 9.8$ )  
3 (A) A, D (B) B, C (C) A, C (D) B, D  
4 48 s from the start  
5 (A)



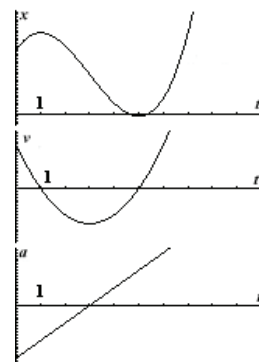
(B)  $5.2857 + 2.4667 + 12.3365 = 20.0889 \text{ s}$  (C) 7.2457 s

- 6  $\frac{8x}{(1-x^2)}$ ;  $0 < x < 1$

- 7  $1\frac{5}{6}$   
8 (A)  $v = 1.5 - 2 \sin(\frac{t}{2})$ ;  $t \geq 0$  (B)  $a = -\cos(\frac{t}{2})$ ;  $t \geq 0$  (C)  $-0.5 \text{ ms}^{-1}$

## 19 – Kinematics and its applications

- 1  $v = \pm \sqrt{24 - 6x^2}$   
2 (A)  $v(t) = 36t^2 - 36t + 7$  for  $t \geq 0$  (B)  $v = -0.19$ ;  $t = 0.26$ ; possible reason: (C) 1  
3 (A) No answer given (B)  $x = 2 \tan(2t + 0.9828)$ ; note  $C = \tan^{-1}(1.5)$   
4  $-2 \sin 2x \cos 2x = -\sin 4x$   
5 (A) (B) No answer given (C)  $x(t) = \frac{1}{2e^{2t} - 1}$   
6 (A) (i)  $v = 3t^2 - 18t + 15$ ;  $t \geq 0$   
(ii)  $a = 6t - 18$ ;  $t \geq 0$   
(B) See graph at right  
(C) 24 m;  $15 \text{ ms}^{-1}$ ;  $-18 \text{ ms}^{-2}$   
(D)  $t = 1 \text{ s}$ ,  $5 \text{ s}$ ;  $x = 31 \text{ m}$ ,  $-1 \text{ m}$   
(E)  $t = 3 \text{ s}$ ;  $x = 15 \text{ m}$   
(F) (i)  $t > 5 \text{ s}$  (ii) never  
(iii)  $0 < t < 1 \text{ s}$  (iv)  $1 < t < 5 \text{ s}$   
7 (A)  $x = \frac{1}{4}t^4 - 2t^3 + \frac{5}{2}t^2 + 3$   
(B)  $3\frac{1}{2} \text{ cm}$



## 20 – Kinematics in real situations

- 1 (A) No answer given (B) 7.9 units  
2 (A)  $D = kv^2$  (B)  $\ddot{x} = g - kv^2$  (C) No answer given (D)  $\frac{g}{2916}$   
(E) (F) No answer given  
3 (A)  $t = \frac{\tan \theta}{\omega}$  (B) (C) No answer given (D) 2.86 cycles per sec

## 21 – Perms, combs and probability

- 1 (A) No answer given (B) 252  
2 (A) 495 (B)  $\frac{1}{495}$   
3 (A)  $2^n - 2$  (B) 56  
4 (A) 243 (B) 60 (C) 320  
5 No answer given  
6 (A) (i)  $q^2 = (1-p)^2$  (ii)  $2p - p^2$   
(B) (i)  $4pq^3 + q^4 = 4p(1-p)^3 + (1-p)^4$  (ii)  $6p^2 - 8p^3 - 3p^4$   
(C)  $0 < p < 0.333$   
7 (A)  $1 - (\Pr(X=0) + \Pr(X=1) + \Pr(X=2))$   
(B)  $1 - (\Pr(X=4) + \Pr(X=5))$  (C)  $1 - (\Pr(X=4) + \Pr(X=5))$   
(D)  $1 - (\Pr(X=0) + \Pr(X=1) + \Pr(X=2))$   
8 (A) 0.410 (B) Possible reason:  $190 \times 9^{18} > 9^{19}$   
(C) More than 29 meetings; 6 years  
9 (A)  $\frac{1}{4}$  (B) No answer given (C) 0.6817 (D) 0.9948

## 22 – Checklist for success

No answers given.

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### Topics include

Some stuff is assumed knowledge  
Best ways factorise  
Good algebra skills are a must  
De Moivre's Theorem  
Complex numbers and algebra  
Functions and relations  
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ABN 50 074 266 023  
108 Main Rd  
Clayton South  
VIC 3169  
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